Research Diary

Lonely Runner Conjecture Research Journal

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NSERC Undergraduate Student Research Award (USRA)

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Summer 2026

Last Modified: June 19, 2026

Contents

Week 1 (May 19)

N.B.: This research journal is maintained by Asfi Hassan to document the progress, ideas, implementation, and findings arising from research conducted under the supervision of Dr. Ahmed Biniaz at the University of Windsor as part of the NSERC Undergraduate Student Research Award (USRA). This project investigates a computational geometry approach to a problem closely related to the Lonely Runner Problem, a well-known open problem in number theory and discrete mathematics. The purpose of this journal is to provide a clear and organized record of my work throughout the Summer 2026 research term. It is intended to document my understanding, progress, and software development rather than to claim original authorship of all ideas presented. Many of the mathematical concepts and techniques discussed are drawn from existing research and are included to support my learning and to show how they inform my investigations and implementations. Entries may include mathematical observations, algorithmic developments, software design and implementation details, computational experiments, visualizations, meeting notes, and reflections on challenges encountered during the project. This journal serves as a comprehensive record of both the research process and the software tools developed to better understand this problem. Artificial intelligence tools were used to assist in understanding advanced mathematical concepts and research papers, and to support the preparation and refinement of this research journal at an undergraduate computer science level.

Foundational Understanding of the Lonely Runner Problem

Review of Related Literature

The Lonely Runner Conjecture (LRC) is a well-known problem in number theory and combinatorics involving runners moving at distinct constant speeds around a unit circle. The conjecture states that for any set of $n+1$ runners, each runner will eventually be at least $\frac{1}{n+1}$ units away from every other runner along the circle. By fixing one runner at speed 0, the problem reduces to finding a time t such that the minimum circular distance between the stationary runner and all moving runners is at least $\frac{1}{n+1}$. In this project, the problem is interpreted geometrically by plotting time on the x-axis and the corresponding distance function on the y-axis, producing periodic piecewise-linear graphs whose peaks are farthest from the stationary runner. This transformation converts an abstract modular arithmetic problem into a visual and computational geometry problem. The resulting graphs reveal repeating patterns and critical points where the minimum distance function reaches local maxima. These visual structures make it easier to identify candidate witness times and compare the behavior of different speed sets.

Several key papers guided this research. Ram Krishna Pandey proved the conjecture for lacunary speed sets using a constructive interval-based argument, while recent works by Matthieu Rosenfeld and by Touch Sungkawichai and Tanupat Trakulthongchai used computer-assisted methods to verify the conjecture for up to thirteen total runners. These papers show that the problem can be reduced to a finite search over integer speed sets using modular sieves, bounded reductions, and exhaustive computation. Additional recent preprints refine these techniques by combining modular arithmetic, combinatorial reasoning, and algorithmic filtering to eliminate large classes of potential counterexamples. Inspired by these ideas, this project develops an interactive Python framework using Tkinter and Matplotlib to visualize the distance functions and explore the algorithmic structure underlying one of the most significant open problems in modern combinatorics and number theory.

Problem Sketch from Week 1

During the first meeting in week 1 of the research term, I studied a series of introductory sketches provided by my supervisor, Dr. Ahmed Biniaz, where he explained the core ideas behind the Lonely Runner Problem and several related concepts. These sketches used the unit circle to illustrate how runners move at different constant speeds and showed that a runner is considered lonely when they are at least 1/(k+1) units away from every other runner, where k+1 is the total number of runners. By fixing one runner with speed 0, the problem becomes easier to analyze geometrically, since the distance between a moving runner and the stationary runner increases up to a maximum of 1/2 of the unit circle and then decreases symmetrically. This geometric interpretation provides an intuitive way to visualize the problem and forms the basis for representing and studying it using computational geometry techniques within a plane.

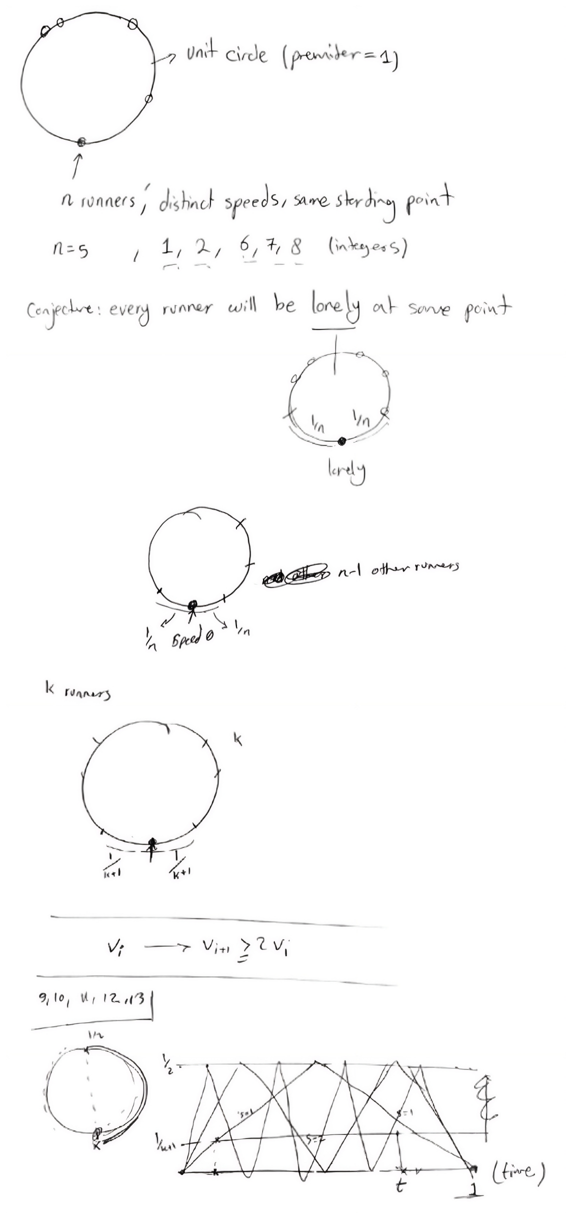

Figure 1: Introductory sketches illustrating the unit circle and graphical interpretation of the Lonely Runner Problem. Sketches provided by Dr. Ahmed Biniaz.

Current Software Implementation

To support the geometric interpretation of the Lonely Runner Conjecture, an initial Python prototype was developed using the `Tkinter` and `Matplotlib` libraries. `Tkinter` is used to create a graphical user interface (GUI), while `Matplotlib` is used to generate and display mathematical plots directly within the application window. The program launches in full-screen mode and provides two input fields: one for the number of speeds, denoted by n, and another for the corresponding integer velocity values entered as a space-separated list.

The core functionality is implemented in the `draw()` function. After validating the user input, the program iterates through each velocity v and computes a sequence of equally spaced isosceles triangles over the interval $[0, 1]$. Each triangle has base length $\frac{1}{v}$ and height $\frac{1}{2}$, representing one period of the distance-to-nearest-integer function associated with that speed. Since the function $\|tv\|$ is periodic and piecewise linear, the repeated triangles provide an exact geometric representation of the runner's relative distance from the stationary runner over time. Different colors are assigned to distinguish the graphs corresponding to different speeds.

The application also includes a `reset()` function, which clears all input fields and removes the current plot, allowing new speed sets to be entered efficiently. The plotting area is embedded directly into the GUI using `FigureCanvasTkAgg`, and the aspect ratio is fixed so that all triangles retain their correct geometric proportions. This prototype establishes the foundational visualization framework for the project and serves as the basis for future extensions, including the computation of intersection points, automatic identification of candidate loneliness times, and algorithmic verification of specific speed configurations.

The core plotting algorithm iterates through each input velocity v and constructs its corresponding distance function. For each speed, the interval $[0, 1]$ is divided into v equal subintervals of length $\frac{1}{v}$. In each subinterval, the program computes the left endpoint, right endpoint, and midpoint, then draws an isosceles triangle of height $\frac{1}{2}$. Each triangle represents one period of the distance-to-nearest-integer function $\|tv\|$. Repeating this process over all subintervals produces the full piecewise-linear graph for that runner, with distinct colors used to differentiate multiple speeds. The code snippet is provided as a figure below:

```
for idx, v in enumerate(values):
    color = colors[idx % len(colors)]
    triangle_base = line_length / v

    for i in range(v):
        base_left_x = i * triangle_base
        base_right_x = (i + 1) * triangle_base
        top_vertex_x = (base_left_x + base_right_x) / 2

        ax.plot(
            [base_left_x, base_right_x, top_vertex_x, base_left_x],
            [0, 0, y, 0],
            color=color
        )
```

Figure 2: Core algorithm above used to generate the triangular representation of the conjecture.

Week 2 (May 26)

Recap

During this weeks discussions with Dr. Ahmed Biniaz, we examined why the Lonely Runner Conjecture becomes significantly more difficult as the number of runners increases. In particular, we discussed the complexity of modern sieving techniques, which use filtering methods to eliminate impossible speed configurations during large computational searches. I was told to take a deeper read into *The Lonely Runner Conjecture turns 60*, to become familiar with concepts. Several special cases of the conjecture were also suggested for independent study and proof attempts, including weaker loneliness bounds and restricted families of runner speeds.

Case 1 — Proving the Weaker Bound $\frac{1}{2k}$

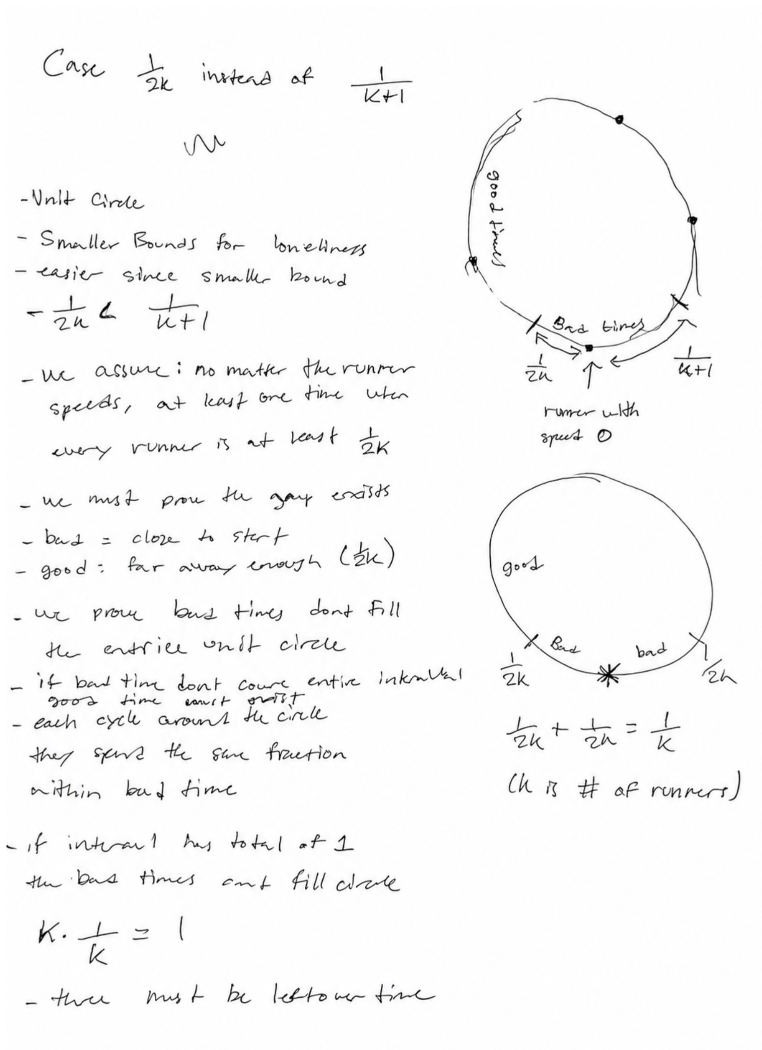

Figure 3: Illustration of the good and bad regions on the unit circle for the 1/(2k) loneliness bound and brainstorming proof for Case 1.

This proof establishes a weaker version of the Lonely Runner Conjecture by showing that there exists a time t such that every runner is at least $\frac{1}{2k}$ away from the starting point. Here, k represents the number of moving runners, and the runner speeds are denoted by $v_1, v_2, \ldots, v_k$.

For each runner v_i, define the set of "bad times" as

$$A_i = \left\{ t \in (0,1) : \|tv_i\| < \frac{1}{2k} \right\}.$$

The idea of "bad times" is simple: these are the times when a runner is still too close to the starting point, meaning the runner is not yet lonely. We create a set A_i to collect all of these bad times together for runner i. The notation $t \in (0, 1)$ means we only examine times between 0 and 1, since the runners move periodically around the unit circle and their positions repeat after one full cycle.

In this expression, t represents time, v_i is the speed of the i-th runner, and $\|tv_i\|$ represents the distance from the runner to the nearest integer position on the unit circle, which corresponds to the distance from the starting point. The condition

$$\|tv_i\| < \frac{1}{2k}$$

therefore means that runner i is closer than $\frac{1}{2k}$ to the starting point, making that time "bad."

For a single runner, the total amount of bad time is less than $\frac{1}{k}$. Since there are k runners, the total amount of bad time across all runners is still less than the entire interval $(0, 1)$. Therefore, there must exist at least one time t which is not contained in any bad set A_i. At this time,

$$\|tv_i\| \geq \frac{1}{2k}$$

for every runner i. Hence, every runner is sufficiently far from the starting point simultaneously.

Weaker Bound $\frac{1}{2k}$, when k = 4

To understand the idea, consider 4 runners. In this case, each runner is considered "bad" when they are within $\frac{1}{8}$ of the starting point. Each runner is only bad for approximately one quarter of the time. At first glance, four runners each contributing one quarter of the interval appears to cover the entire timeline. However, many of these bad intervals overlap, meaning the same moments are counted multiple times. As a result, the bad times cannot cover the entire interval $(0, 1)$. Therefore, there must exist at least one time where none of the runners are bad, implying that all four runners are simultaneously at least $\frac{1}{8}$ away from the starting point.

This does not prove the full Lonely Runner Conjecture, which requires a distance of $\frac{1}{k+1}$, but it does establish the existence of a guaranteed weaker loneliness distance of $\frac{1}{2k}$.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 5: "Gap of loneliness." The weaker bound $\kappa(n) \geq \frac{1}{2n}$ is attributed to Wills.

Case 2 — Speeds Growing by a Factor of Two

This case proves the Lonely Runner Conjecture when the runner speeds increase rapidly. Specifically, the speeds satisfy

$$v_{i+1} \geq 2v_i.$$

This means each runner moves at least twice as fast as the previous runner. For example, possible speeds could be $1, 2, 4, 8, 16$. Such sequences are known as lacunary sequences.

The condition $v_{i+1} \geq 2v_i$ is what makes the proof work. Since each runner is moving at least twice as fast as the previous runner, the corresponding distance function repeats more frequently over the interval $[0, 1]$. Intuitively, this creates many opportunities to find times where additional runners satisfy the loneliness condition without eliminating all previously valid times.

The goal is to show that there exists a time t where every runner is at least

$$\frac{1}{k+1}$$

away from the starting point, where k is the number of moving runners.

The proof works by constructing nested intervals of valid times. An interval is simply a continuous range of times. For example, the interval $[0.2, 0.4]$ contains every time between 0.2 and 0.4. The notation $I_2 \subseteq I_1$ means that I_2 is completely contained inside I_1.

First, an interval I_1 is chosen where the first runner is sufficiently far from the starting point. Then a smaller interval $I_2 \subseteq I_1$ is selected where both the first and second runners satisfy the loneliness condition. This process continues:

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_k.$$

Because each speed is at least twice as large as the previous one, the distance patterns repeat more frequently, creating enough opportunities to continue choosing smaller valid intervals. Since every interval is contained in the previous one, any time remaining in the final interval automatically works for all runners considered so far.

At the end of the construction, the final interval I_k is still nonempty, meaning it contains at least one valid time t. Therefore,

$$\|tv_i\| \geq \frac{1}{k+1}$$

for every runner i, and all runners become lonely simultaneously.

Reference: Ram Krishna Pandey, *A Note on the Lonely Runner Conjecture.* Also summarized in Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 7.3: "Lacunary sequences."

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Case 3 — When $\frac{v_k}{v_1} \leq k$

This case proves the Lonely Runner Conjecture when the fastest runner is not excessively faster than the slowest runner. The condition is

$$\frac{v_k}{v_1} \leq k,$$

where v_1 is the smallest speed, v_k is the largest speed, and k is the number of moving runners. Equivalently,

$$v_k \leq kv_1.$$

This means the fastest runner is at most k times faster than the slowest runner. For example, if there are 4 runners, then the fastest runner can be at most 4 times faster than the slowest runner. The speeds are therefore relatively close together and do not vary too dramatically.

The importance of this condition is that the runners move at comparable rates. If one runner were significantly faster than all the others, the distance patterns would be more difficult to analyze. By keeping the speeds within a controlled range, the structure of the problem becomes much more predictable.

The proof begins by defining

$$N = v_1 + v_k$$

Here, N is simply the sum of the smallest and largest speeds. Under the assumption above, all runner speeds lie inside a controlled interval between

$$\frac{N}{k+1} \quad \text{and} \quad N - \frac{N}{k+1}.$$

In other words, every speed is guaranteed to lie within a specific range determined by the slowest and fastest runners. This prevents any speed from being unusually small or unusually large relative to the others.

Because the speeds are confined to this range, the resulting distance functions exhibit a more regular structure. This allows the authors to apply known results showing that there exists a time t such that

$$\|tv_i\| \geq \frac{1}{k+1}$$

for every runner i. Therefore, all runners become lonely simultaneously.

In simple terms, the proof shows that when the runner speeds are reasonably close together, the system is sufficiently well-behaved that the Lonely Runner Conjecture can be established directly.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 7.1: "Simple remarks and further reductions." "'

Week 3 (June 2)

Recap

During this Week 3 meeting, Dr. Ahmed Biniaz walked me through the three special cases of the Lonely Runner Conjecture that I have been studying. We discussed the intuition behind each proof, focusing on why the arguments work and how they relate to the overall conjecture. We also revisited Case 3, where the fastest runner is not excessively faster than the slowest runner, and clarified several details of the proof. In particular, Dr. Biniaz explained how choosing the time

$$g = \frac{1}{v_1 + v_k}$$

and substituting it into the distance function $\|gv_i\|$ helps show that all runners remain sufficiently far from the starting point. This provided a clearer understanding of why the condition $\frac{v_k}{v_1} \leq k$ guarantees the loneliness requirement and how the distance function is used within the proof.

Dr. Biniaz suggested keeping the definition

$$\|x\| = \min\{x - \lfloor x \rfloor,\ \lceil x \rceil - x\}$$

in mind when studying the conjecture, since it represents the distance from a real number to the nearest integer. This notation appears throughout the Lonely Runner literature and is fundamental to expressing the loneliness condition. Understanding this definition should help provide additional intuition for Case 3 and the role of the distance function in the conjecture.

For the coming week, I will continue reading the relevant papers and studying additional cases, proofs, reductions, and properties related to the Lonely Runner Conjecture. My goal is to strengthen my understanding of the existing literature and develop a clearer picture of the different techniques that have been used to prove special cases of the conjecture.

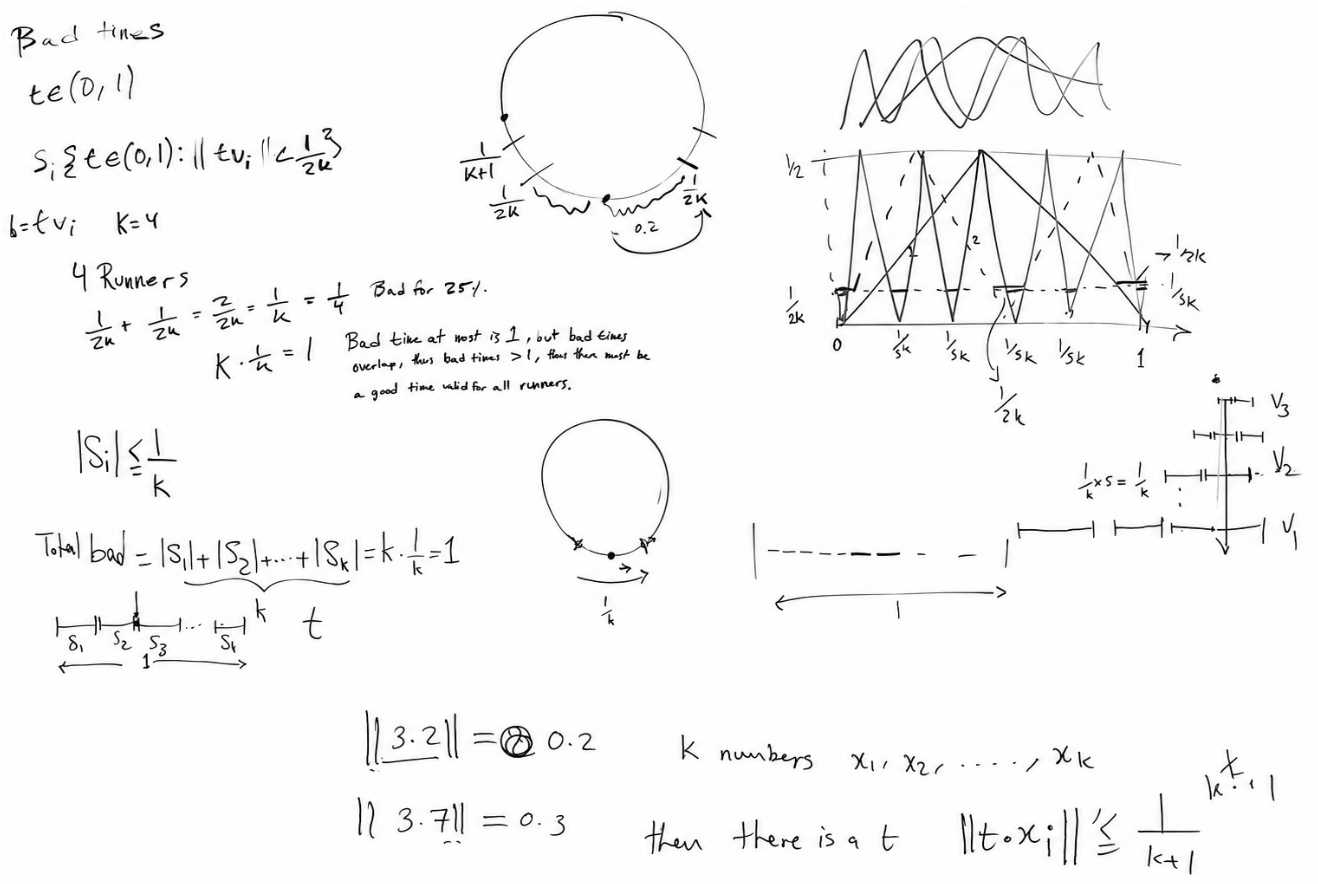

Figure 4: Sketches from meeting illustrating the main ideas behind Cases 1 and 2 of the Lonely Runner Conjecture, including bad times and runner separation on the unit circle.

Understanding the Distance Function $\|x\|$

The notation $\|x\|$ appears throughout the Lonely Runner literature and represents the distance from a real number x to the nearest integer. It is defined by

$$\|x\| = \min\{x - \lfloor x \rfloor,\ \lceil x \rceil - x\}.$$

To understand this formula, it is helpful to examine each symbol individually. The notation $\lfloor x \rfloor$ means the greatest integer less than or equal to x, often described as rounding x down. Similarly, $\lceil x \rceil$ means the smallest integer greater than or equal to x, often described as rounding x up. The notation $\min\{\cdot\}$ means that we select the smaller of the two values inside the braces.

The expression $x - \lfloor x \rfloor$ measures the distance from x to the integer immediately below it, while $\lceil x \rceil - x$ measures the distance from x to the integer immediately above it. Since $\|x\|$ represents the distance to the nearest integer, we simply take whichever of these two distances is smaller.

For example, if $x = 3.2$, then $\lfloor x \rfloor = 3$ and $\lceil x \rceil = 4$. The distance to 3 is 0.2, while the distance to 4 is 0.8. Since 0.2 is smaller, we obtain $\|3.2\| = 0.2$. Similarly, if $x = 3.8$, the distance to 3 is 0.8 and the distance to 4 is 0.2, so $\|3.8\| = 0.2$.

In the context of the Lonely Runner Conjecture, the quantity $\|tv_i\|$ measures how far runner i is from the starting point at time t. This distance function appears throughout the literature and is fundamental to expressing the loneliness condition. Understanding this notation provides important intuition for many proofs and reductions related to the conjecture.

Case 4 — Proof for 2 runners, where $n = 2$

For the case $n = 2$, the Lonely Runner Conjecture asks us to prove that

$$\kappa(2) = \frac{1}{3}.$$

Here, $n = 2$ means there are two moving speeds, usually written as

$$V = \{v_1, v_2\},$$

where $v_1 < v_2$. The notation $\kappa(V)$ represents the largest guaranteed loneliness distance for the speed set V. Therefore, proving $\kappa(2) = \frac{1}{3}$ means that for any two speeds, there is always some time when both moving runners are at least $\frac{1}{3}$ of a lap away from the stationary runner at the origin.

The main challenge is determining which time t to examine. Since time can take infinitely many values, checking every possible time individually would be impossible. The proof therefore looks for a structured family of candidate times that are likely to produce large distances from the starting point.

One important idea used in this proof is that the best loneliness time can be chosen in the form

$$t_0 = \frac{\ell}{v_1 + v_2},$$

where ℓ is a positive integer. At first glance, this formula may seem unusual, but its purpose is to balance the motion of the two runners. The quantity $v_1 + v_2$ is the combined speed of both runners, and using it in the denominator creates times that naturally align with the movement patterns of both runners simultaneously.

The symbol ℓ simply selects which candidate time we are testing. For example, the proof considers times such as

$$\frac{1}{v_1 + v_2}, \qquad \frac{2}{v_1 + v_2}, \qquad \frac{3}{v_1 + v_2},$$

and so on. Instead of searching through infinitely many real numbers, the proof only needs to study these special candidate times.

At such a time, the distance from runner i to the origin is measured using

$$\|t_0 v_i\|.$$

Here, $t_0 v_i$ represents how far runner i has travelled after time t_0. The notation $\|t_0 v_i\|$ means the distance from that position to the nearest integer. Since every integer corresponds to completing a whole number of laps and returning to the starting point, $\|t_0 v_i\|$ is exactly the distance from the runner to the starting point on the unit circle.

The proof then shows that one can choose a value of ℓ that places both runners as far from the starting point as possible. This leads to the bound

$$\kappa(V) = \frac{\lfloor (v_1 + v_2)/2 \rfloor}{v_1 + v_2}.$$

The symbol $\lfloor x \rfloor$ means rounding x down to the nearest integer. The quantity

$$\lfloor (v_1 + v_2)/2 \rfloor$$

therefore represents the largest whole number that does not exceed half of $v_1 + v_2$. Intuitively, this corresponds to placing the runners as close as possible to the halfway point around the circle, which maximizes their distance from the origin.

This fraction is always at least

$$\frac{1}{3},$$

which implies

$$\kappa(V) \geq \frac{1}{3}.$$

Therefore, for any two speeds v_1 and v_2, there exists a time when both runners are at least $\frac{1}{3}$ of the track away from the origin.

In simple terms, the proof shows that with only two moving runners, their positions can always be balanced so that neither runner is too close to the starting point. This is why the Lonely Runner Conjecture is fully true for $n = 2$.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 6.1: "Case $n = 2$."

Invisible Runners

The *Invisible Runner Problem* is a variation of the Lonely Runner Conjecture introduced by Czerwinski and Grytczuk. The main idea is simple: instead of requiring every runner to satisfy the loneliness condition, one runner is allowed to be ignored, or made "invisible." The question then becomes whether the remaining runners can still achieve the required separation.

Suppose we have a set of runner speeds

$$V = \{v_1, v_2, \ldots, v_n\},$$

where n is the number of runners and each v_i represents a speed. In the original Lonely Runner Conjecture, every runner must eventually become lonely. The Invisible Runner Problem asks what happens if we remove one runner from consideration and focus only on the remaining runners.

The main result states that for every set of n runner speeds, there is always at least one runner whose removal guarantees a loneliness distance of at least

$$\frac{1}{n}.$$

In mathematical notation, this is written as

$$\kappa(V \setminus \{v\}) \geq \frac{1}{n}.$$

Here, $V \setminus \{v\}$ means the speed set obtained after removing runner v, and κ represents the guaranteed loneliness distance. In simple terms, the theorem says that every speed set contains at least one runner whose removal makes the remaining problem easier.

The intuition behind the proof comes from averaging. Imagine observing all runners over time and counting how many runners are too close to the starting point. The proof shows that if the guaranteed loneliness distance were smaller than $\frac{1}{n}$, then the average number of runners that are too close to the origin would be less than 2.

This is important because if there were always at least two runners too close to the starting point, then the average would have to be at least 2. Since the average is actually less than 2, there must be some moment when there is at most one runner that is too close to the origin.

For example, suppose there are five runners. At a particular time, it might happen that four runners are already far enough from the starting point, while only one runner remains too close. At that moment, the loneliness condition is almost satisfied. The only obstacle is the single runner that is still too close to the origin.

This observation is the key idea behind the Invisible Runner Problem. If we simply remove the one runner causing the problem and make that runner "invisible," then all of the remaining runners already satisfy the required distance of at least $\frac{1}{n}$ from the starting point. Therefore, there must always exist at least one runner whose removal guarantees the loneliness condition for the rest of the group.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 10.1: "Invisible runners."

Week 4 (June 9)

Recap

During this weeks meeting, Dr. Ahmed Biniaz and I continued discussing several important topics related to the Lonely Runner Conjecture. A significant portion of the meeting focused on the *Invisible Runner Problem*, where we carefully went through the proof and developed intuition for why removing a single runner can guarantee the loneliness condition for the remaining runners. In particular, we discussed the averaging argument used in the proof and why assuming

$$\kappa(V) < \frac{1}{n}$$

implies that the average number of runners inside the bad region is

$$2\kappa n < 2.$$

This means that, on average, fewer than two runners are too close to the starting point, so there must exist some moment where at most one runner is violating the loneliness condition.

We also explored the *Billiard Ball Trajectory* interpretation of the conjecture. Rather than directly studying runners moving around a circular track, Dr. Biniaz introduced an alternative visualization by flattening the circular motion into a linear representation. In this interpretation, the horizontal axis represents time t, while the vertical axis represents distance or completed laps d. The shaded intervals represent "bad" regions, meaning positions where a runner is too close to the starting point and therefore does not satisfy the loneliness condition. This provided an intuitive geometric way of understanding how runner positions evolve over time and how valid loneliness times can be identified.

For the coming week, I will continue strengthening my understanding of Cases 1–3, while also continuing to study the *Invisible Runner Problem*. I plan to leave the *Billiard Ball Trajectory* and *View-Obstruction* interpretation for later, since these currently appear to be the most difficult concepts. These four topics will be my primary focus moving forward.

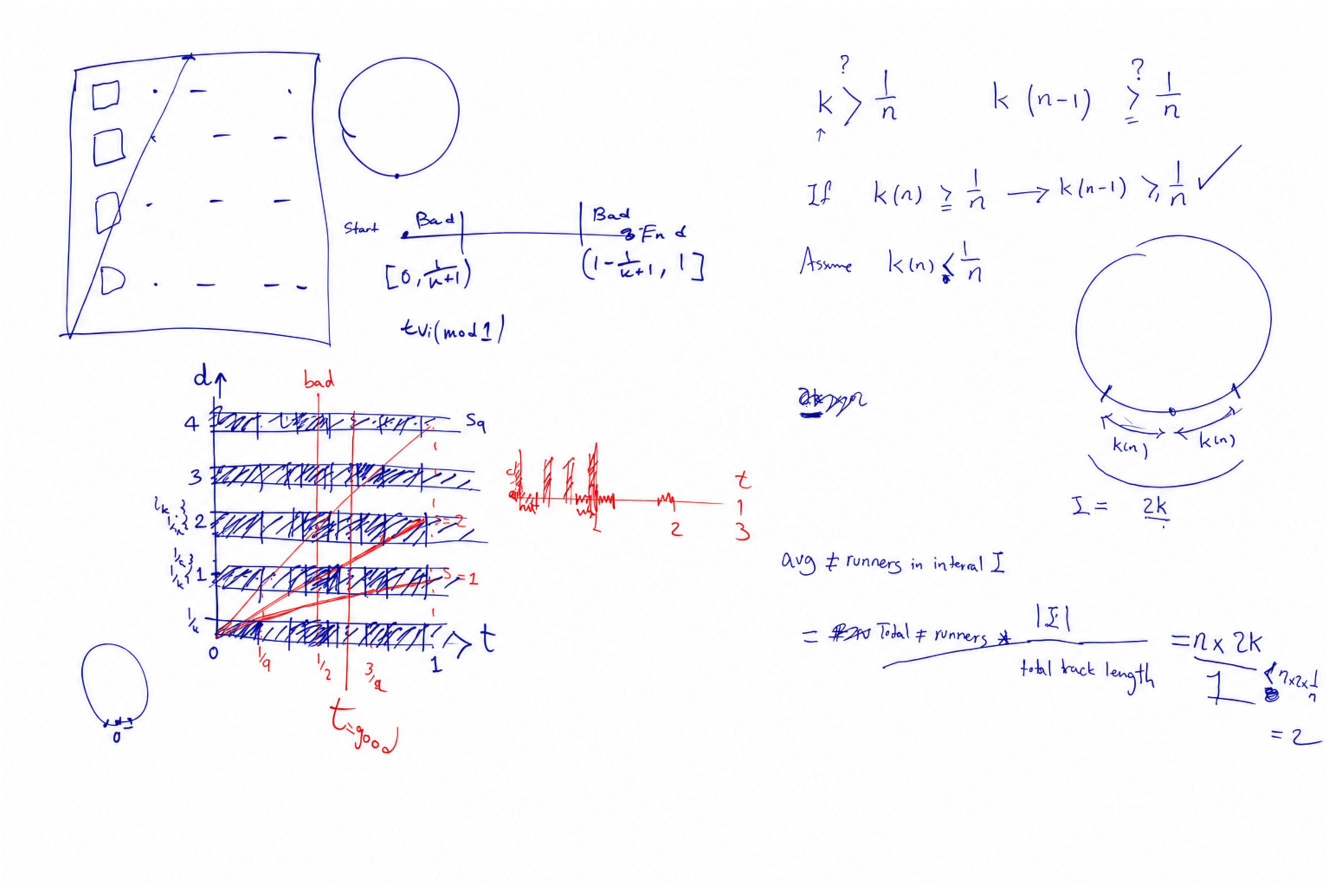

Figure 5: Visual representations of the Lonely Runner Conjecture showing averaging arguments, bad regions, and geometric trajectory interpretations.

Case 1 — Deeper Dive $\frac{1}{2k}$

To better understand the weaker bound proof, it is helpful to study more carefully why the argument works and how each part of the proof is constructed. The goal is to show that for any set of k moving runners, there always exists some time where every runner is at least

$$\frac{1}{2k}$$

away from the starting point.

In this setting, k represents the number of moving runners. The quantity

$$\kappa(k)$$

represents the largest guaranteed loneliness distance that can always be achieved for any possible set of k runner speeds. Therefore, proving

$$\kappa(k) \geq \frac{1}{2k}$$

means that no matter what speeds are chosen, there is always at least one time where all runners are simultaneously at least $\frac{1}{2k}$ of a lap away from the origin.

The proof begins by shifting focus away from searching directly for a good time. Since time is continuous, checking every possible value of t would be impossible. Instead, the proof studies the opposite idea by looking at the times when runners are *too close* to the starting point.

For each runner with speed v_i, define the set

$$A_i = \{t \in (0,1) : \|tv_i\| < \frac{1}{2k}\}.$$

The variable t represents time, and we only examine the interval

$$0 < t < 1$$

because runner positions repeat after one complete cycle. The value v_i is the speed of runner i, and the expression

$$tv_i$$

tells us how far the runner has travelled after time t.

The notation

$$\|tv_i\|$$

represents the distance from that position to the nearest integer. Since integers correspond to completing a full lap and returning to the starting point, this quantity measures how far the runner currently is from the origin on the unit circle.

The set A_i therefore represents all *bad times* for runner i. These are exactly the times when the runner is still too close to the starting point and has not yet reached the required loneliness distance.

To understand how large this bad-time set is, imagine the track has total length

$$1.$$

A runner is considered bad whenever it lies within distance

$$\frac{1}{2k}$$

from the starting point.

Since the starting point can be approached from both sides of the circle, there are two bad regions.

The first bad region is

$$\left[0, \frac{1}{2k}\right)$$

and the second bad region is

$$\left(1 - \frac{1}{2k}, 1\right].$$

Each region has length

$$\frac{1}{2k}.$$

Adding both together gives total bad length

$$\frac{1}{2k} + \frac{1}{2k} = \frac{1}{k}.$$

This means that each runner spends exactly

$$\frac{1}{k}$$

of the total time interval inside bad positions.

Now consider all k runners together. Since each runner contributes at most

$$\frac{1}{k}$$

of bad time, the total amount of bad time contributed by all runners is at most

$$k \times \frac{1}{k} = 1.$$

At first glance, this seems to cover the entire interval.

However, this does not mean the entire timeline is actually covered. The reason is that many runners can be bad at the same moment, meaning their bad intervals overlap. When intervals overlap, the same portion of time is counted multiple times.

Because of these overlaps, the bad times cannot completely cover every possible time in the interval

$$(0, 1).$$

Therefore, there must exist at least one time which does not belong to any bad-time set A_i.

At that special time, every runner satisfies

$$\|tv_i\| \geq \frac{1}{2k}$$

for every runner i.

This means every runner is simultaneously far enough away from the starting point.

Therefore, we conclude that

$$\kappa(k) \geq \frac{1}{2k}.$$

A useful way to understand the proof is to imagine trying to cover the entire time interval using all bad times from every runner. Even though each runner contributes some bad intervals, many of these intervals overlap with each other. Since overlapping means time is counted repeatedly, the bad intervals cannot completely block every moment. This guarantees that at least one good time exists where every runner satisfies the required distance.

This result does not prove the full Lonely Runner Conjecture, which requires distance

$$\frac{1}{k+1},$$

but it does establish an important weaker bound which became one of the earliest general results for the conjecture and served as the foundation for many later improvements in the literature.

References: Jörg Wills (1967, 1968); Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 5: "Gap of loneliness."

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Can We Do Better Than $\frac{1}{2n}$?

After proving the weaker bound

$$\kappa(n) \geq \frac{1}{2n},$$

a natural question arises: can this guaranteed distance be improved further?

To understand this question, recall that n represents the number of moving runners, and the quantity

$$\kappa(n)$$

represents the largest distance that can always be guaranteed for every possible set of n runner speeds. In simple terms, $\kappa(n)$ tells us the smallest loneliness distance we know every speed set must achieve. The weaker bound proved by Wills guarantees that every runner can eventually become at least

$$\frac{1}{2n}$$

away from the starting point. However, the actual Lonely Runner Conjecture asks for the stronger bound

$$\frac{1}{n+1}.$$

Since

$$\frac{1}{n+1} > \frac{1}{2n},$$

the weaker bound does not fully solve the conjecture. This led mathematicians to ask whether the value

$$\frac{1}{2n}$$

could be gradually pushed closer to

$$\frac{1}{n+1}.$$

The original proof used an idea from probability. For each runner with speed v_i, we defined the set

$$A_i = \left\{ t \in (0,1) : \|tv_i\| < \frac{1}{2n} \right\}.$$

The set A_i represents all times when runner i is too close to the starting point. These are called bad times because the runner is not lonely during these moments.

The proof then studied the expression

$$Pr(A_1 \cup A_2 \cup \cdots \cup A_n).$$

This notation needs to be understood carefully. The symbol

$$Pr$$

stands for *probability*. Here, probability simply means the fraction of time something happens while looking at all possible times between 0 and 1. The symbol

$$\cup$$

means *union*. In simple terms, union means combining several sets together. So the expression

$$A_1 \cup A_2 \cup \cdots \cup A_n$$

means the collection of all times when *at least one runner* is bad. Therefore,

$$Pr(A_1 \cup A_2 \cup \cdots \cup A_n)$$

represents the fraction of the total timeline where at least one runner is too close to the origin. The proof then uses an inequality called the *union bound*

$$Pr(A_1 \cup A_2 \cup \cdots \cup A_n) \leq Pr(A_1) + Pr(A_2) + \cdots + Pr(A_n).$$

The symbol

$$\leq$$

means "less than or equal to." This formula says that the probability of the combined bad set can never be larger than the sum of the individual bad probabilities. To understand why, imagine two runners. Suppose runner 1 is bad during 20% of the time, and runner 2 is also bad during 20% of the time. Naively adding them gives

$$20\% + 20\% = 40\%.$$

However, if both runners are bad at the same moment for part of the time, then some moments have been counted twice. For example, if both runners are simultaneously bad during 10% of the timeline, then the true combined bad time is only

$$30\%.$$

This means simply adding probabilities overestimates how much bad time actually exists. This is exactly the weakness of the original proof. The union bound assumes the worst case by adding everything together without carefully checking whether bad intervals overlap. Later mathematicians realized that this method is not precise enough. Instead of simply adding bad intervals, they asked a better question:

How often are multiple runners bad at the same time?

To study this, researchers introduced expressions such as

$$Pr(A_i \cap A_j).$$

Here the symbol

$$\cap$$

means *intersection*. Intersection means finding the part that two sets share in common. So

$$A_i \cap A_j$$

represents the set of times when both runner i and runner j are simultaneously bad. Therefore,

$$Pr(A_i \cap A_j)$$

means the fraction of time where both runners are too close to the starting point at exactly the same moment. This gives more information than the original proof. Instead of only knowing how much bad time each runner contributes individually, we now know how much bad time overlaps. If many bad intervals overlap, then the total bad time is smaller than the union bound originally predicted. This led Chen to improve the weaker bound. Chen proved

$$\kappa(n) \geq \frac{1}{2n - 1 - \frac{1}{2n-3}}.$$

At first glance this formula looks complicated, but its purpose is simple. It guarantees a distance slightly larger than

$$\frac{1}{2n}.$$

To understand why this is better, compare an example. Suppose

$$n = 5.$$

The original weaker bound gives

$$\kappa(5) \geq \frac{1}{10}.$$

This means we guarantee a distance of at least

$$0.10.$$

Chen's formula gives approximately

$$\kappa(5) \geq 0.113.$$

This number is larger. A larger lower bound is always stronger because it means we can guarantee runners get farther away from the starting point. In simple terms, Chen improved the proof by counting bad times more carefully. Later, Chen and Cusick introduced a number theoretic approach involving prime numbers. They proved that for positive integers a and prime numbers p,

$$\kappa(n) \geq \frac{a}{p}.$$

A prime number is a number divisible only by 1 and itself, such as

$$2, 3, 5, 7, 11.$$

The purpose of this formula is to exploit special arithmetic structure between runner speeds. Their idea was that certain prime number relationships force runner positions to spread out more evenly around the circle. Under special conditions, this gives a stronger guarantee than the original weaker bound. For example, when

$$2n - 3$$

is prime, they prove

$$\kappa(n) \geq \frac{1}{2n-3}.$$

Suppose again that

$$n = 5.$$

Then

$$2n - 3 = 7.$$

Since 7 is prime, this gives

$$\kappa(5) \geq \frac{1}{7}.$$

Numerically this becomes

$$\kappa(5) \geq 0.143.$$

Compare this with the original weaker bound

$$0.10.$$

Since

$$0.143 > 0.10,$$

this is a stronger result. Researchers then continued improving the union bound itself. Instead of only studying individual bad sets and pairwise overlaps, they used a more advanced probability tool called a Bonferroni inequality. Wills' original proof is simple because it roughly counts bad times. Later researchers realized this counting method is not precise enough, so they began studying overlaps between bad intervals, prime number structure, and deeper number theoretic patterns. Even after decades of work, mathematicians have only managed small improvements. This shows that while proving

$$\kappa(n) \geq \frac{1}{2n}$$

is relatively straightforward, pushing this value closer to the full conjectured bound

$$\frac{1}{n+1}$$

remains one of the hardest parts of studying the Lonely Runner Conjecture.

References: Jörg Wills (1967, 1968); Chen; Chen and Cusick; Terence Tao; Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 5: "Gap of loneliness." "'

Case 2 — Deeper Dive into Speeds Growing by a Factor of Two

This case studies a special family of runner speeds where each speed is at least twice as large as the previous one. In mathematical notation, this condition is written as

$$v_{i+1} \geq 2v_i.$$

Here, v_i represents the speed of the i-th runner, and v_{i+1} represents the next speed in the ordered list. The inequality means that every new runner is at least twice as fast as the runner before it. For example, the speeds

$$1, 2, 4, 8, 16$$

satisfy this condition because each number is at least double the previous one. These types of speed sequences are called *lacunary sequences*. In simple terms, a lacunary sequence is a sequence that grows very quickly and has large gaps between consecutive terms.

The goal is to prove that the Lonely Runner Conjecture holds for these rapidly growing speeds. If there are k moving runners, then the conjecture asks us to find a time t such that every runner is at least

$$\frac{1}{k+1}$$

away from the starting point. This distance is the full conjectured loneliness bound. Therefore, unlike the weaker bound $\frac{1}{2k}$, this case proves the actual Lonely Runner Conjecture for this special type of speed set.

The reason the condition $v_{i+1} \geq 2v_i$ is useful is that faster runners repeat their distance pattern more often over the time interval $[0, 1]$. A runner with speed v completes v full laps during this interval, so its distance-to-starting-point graph repeats v times. Therefore, if one runner has speed 1, another has speed 2, and another has speed 4, the faster runners create more repeated opportunities to be far from the starting point.

The proof uses the idea of *nested intervals*. An interval is simply a continuous range of times. For example,

$$[0.2, 0.4]$$

means every time between 0.2 and 0.4, including the endpoints. In this proof, an interval represents a collection of times that are still possible candidates for satisfying the loneliness condition.

The proof starts by finding an interval I_1 of times where the first runner is far enough from the starting point. Then, inside I_1, the proof finds a smaller interval I_2 where both the first and second runners are far enough from the starting point. Then, inside I_2, it finds an even smaller interval I_3 where the first three runners are all far enough from the starting point. This process continues until all k runners have been included.

This is written as

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_k.$$

The symbol $\supset$ means "contains." Therefore, $I_1 \supset I_2$ means that I_2 is completely inside I_1. The purpose of this nesting is important: once we choose I_2 inside I_1, every time in I_2 automatically still works for the first runner. Similarly, every time in I_3 automatically works for the first two runners, because I_3 is inside I_2. By the end, any time inside I_k works for all k runners.

The main reason this construction works is the fast growth of the speeds. Since each new speed is at least twice the previous one, the next runner's distance function repeats many times inside the interval already chosen for the earlier runners. This gives the proof enough room to select a smaller interval that still lies inside the previous one while also satisfying the next runner's loneliness condition.

In the paper, this is summarized by Theorem 18, which states that for every $n \geq 2$, if a speed set V satisfies

$$v_{i+1} \geq 2v_i \qquad \text{for } i \in [n-1],$$

then

$$\kappa(V) \geq \frac{1}{n+1}.$$

Here, n is the number of moving runners, V is the set of speeds, and $[n-1]$ means the set of indices

$$\{1, 2, \ldots, n-1\}.$$

The notation $\kappa(V)$ represents the best guaranteed loneliness distance for the speed set V. Therefore, the inequality

$$\kappa(V) \geq \frac{1}{n+1}$$

means that the speed set V satisfies the Lonely Runner Conjecture.

A simple example is

$$V = \{1, 2, 4\}.$$

Here, there are $n = 3$ moving runners. The required loneliness distance is therefore

$$\frac{1}{n+1} = \frac{1}{4}.$$

The speeds satisfy the doubling condition because

$$2 \geq 2(1)$$

and

$$4 \geq 2(2).$$

So Theorem 18 guarantees that there exists a time t where all three runners are at least $\frac{1}{4}$ away from the starting point.

The paper also discusses broader versions of this idea. A sequence (v_n) is called ϵ-lacunary if there exists some $\epsilon > 0$ such that

$$v_{n+1} \geq (1+\epsilon)v_n$$

for all $n \geq 1$. This means each speed is larger than the previous one by at least a fixed multiplicative amount. The doubling case is a special example where the growth factor is 2. In that case,

$$1 + \epsilon = 2,$$

so

$$\epsilon = 1.$$

The purpose of this more general definition is to study speed sequences that still grow quickly, even if they do not exactly double each time.

Later results weaken the doubling requirement even further. The paper states that for sufficiently large n, it is enough to assume

$$v_{i+1} \geq \left(1 + \frac{22 \log n}{n}\right) v_i.$$

This condition is less strict than requiring $v_{i+1} \geq 2v_i$. The expression

$$1 + \frac{22 \log n}{n}$$

is a growth factor. It says that each speed only needs to be larger than the previous speed by a certain percentage depending on n. When n is large, the quantity $\frac{22 \log n}{n}$ becomes relatively small, so the required growth is much weaker than doubling. This shows that researchers have improved the original lacunary condition and can prove the conjecture under less restrictive assumptions.

In simple terms, Case 2 works because rapidly increasing speeds create many repeated distance patterns. These repeated patterns make it possible to keep narrowing down the set of valid times without losing all possible choices. The nested intervals are the mechanism that keeps track of this process: each interval preserves the good times for the previous runners while adding the next runner into the condition.

References: Ram Krishna Pandey, *A Note on the Lonely Runner Conjecture*; Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 7.3: "Lacunary sequences."

Case 3 — Deeper Dive into $\frac{v_k}{v_1} \leq k$

In this case, we study the situation where the fastest runner is not too much faster than the slowest runner. Suppose the moving runner speeds are ordered as

$$v_1 \leq v_2 \leq \cdots \leq v_k,$$

where v_1 is the slowest speed, v_k is the fastest speed, and k is the number of moving runners. The main assumption for this case is

$$\frac{v_k}{v_1} \leq k.$$

This means that the fastest runner is at most k times faster than the slowest runner. Equivalently, we can multiply both sides by v_1 to get

$$v_k \leq kv_1.$$

Since every speed v_i lies between the slowest and fastest speeds, we know

$$v_1 \leq v_i \leq v_k.$$

Using the condition $v_k \leq kv_1$, this becomes

$$v_1 \leq v_i \leq kv_1.$$

So every runner speed is trapped inside the controlled interval from v_1 to kv_1. This is the key reason this case is easier: no runner is extremely faster than the others.

Now we choose a specific time

$$g = \frac{1}{v_1(k+1)}.$$

This is called a good time because it is chosen using the slowest speed v_1 and the target loneliness distance $\frac{1}{k+1}$. The reason for choosing this value is that when the slowest runner runs for time g, its position is

$$v_1 g = v_1 \left(\frac{1}{v_1(k+1)} \right) = \frac{1}{k+1}.$$

So the slowest runner is exactly $\frac{1}{k+1}$ away from the starting point at time g. This matches the Lonely Runner Conjecture distance.

For any runner i, its position at time g is

$$v_i g.$$

Substituting the value of g, we get

$$v_i g = v_i \left(\frac{1}{v_1(k+1)} \right) = \frac{v_i}{v_1(k+1)}.$$

The distance from runner i to the nearest integer is therefore

$$\|v_i g\| = \left\| \frac{v_i}{v_1(k+1)} \right\|.$$

Now we use the speed bounds. Since

$$v_1 \leq v_i \leq k v_1,$$

we divide everything by $v_1(k+1)$. This gives

$$\frac{v_1}{v_1(k+1)} \leq \frac{v_i}{v_1(k+1)} \leq \frac{k v_1}{v_1(k+1)}.$$

After simplifying, we get

$$\frac{1}{k+1} \leq \frac{v_i}{v_1(k+1)} \leq \frac{k}{k+1}.$$

So at the chosen time g, every runner's position lies between

$$\frac{1}{k+1} \quad \text{and} \quad \frac{k}{k+1}.$$

This interval is important because every point inside it is at least $\frac{1}{k+1}$ away from an integer. The nearest integers are 0 and 1. A point in this interval is not too close to 0, because it is at least $\frac{1}{k+1}$. It is also not too close to 1, because the largest possible value is $\frac{k}{k+1}$, whose distance from 1 is

$$1 - \frac{k}{k+1} = \frac{1}{k+1}.$$

Therefore,

$$\left\| \frac{v_i}{v_1(k+1)} \right\| \geq \frac{1}{k+1}.$$

Since

$$\|v_i g\| = \left\| \frac{v_i}{v_1(k+1)} \right\|,$$

we conclude that

$$\|v_i g\| \geq \frac{1}{k+1}$$

for every runner i.

This proves that at the same time

$$g = \frac{1}{v_1(k+1)},$$

all runners are at least $\frac{1}{k+1}$ away from the starting point. Therefore, all runners are lonely simultaneously, and the Lonely Runner Conjecture holds in this case.

In simple terms, this case works because the speeds are close enough together. By choosing the time g based on the slowest runner, the slowest runner lands exactly at distance $\frac{1}{k+1}$, while the fastest runner can go no farther than $\frac{k}{k+1}$. Since all other runners have speeds between these two, they also land inside the safe interval

$$\left[\frac{1}{k+1}, \frac{k}{k+1} \right].$$

Thus, no runner is too close to the starting point at time g.

Week 5 (June 16)

Recap

As stated last week, I will continue strengthening my understanding of Cases 1–3, while also continuing to study the *Invisible Runner Problem*. I plan to leave the *Billiard Ball Trajectory* and *View-Obstruction* interpretation for later, since these currently appear to be the most difficult concepts. These four topics will be my primary focus moving forward.

Deeper Dive into the Invisible Runner

The *Invisible Runner Problem* is a variation of the Lonely Runner Conjecture. Instead of requiring every runner to become lonely at the same time, we are allowed to remove one or more runners from the set. These removed runners are called "invisible" because we ignore them when checking the loneliness condition.

Suppose we have a set of n runner speeds

$$V = \{v_1, v_2, \ldots, v_n\}.$$

Here, V is the full speed set, n is the number of runners, and each v_i is one runner's speed. If we remove one runner $v \in V$, then the remaining set is written as

$$V \setminus \{v\}.$$

The symbol $\setminus$ means "remove from the set." So $V \setminus \{v\}$ means the original speed set after removing the runner with speed v.

The main invisible runner result says that for every set V of n positive integer speeds, there exists some runner $v \in V$ such that

$$\kappa(V \setminus \{v\}) \geq \frac{1}{n}.$$

This means that after removing one carefully chosen runner, the remaining $n-1$ runners have guaranteed loneliness distance at least $\frac{1}{n}$. The number $\frac{1}{n}$ comes from the usual Lonely Runner distance. Since removing one runner leaves $n-1$ runners, the expected distance becomes

$$\frac{1}{(n-1)+1} = \frac{1}{n}.$$

So $\frac{1}{n}$ is exactly the correct target distance for the remaining runners.

The proof uses averaging. For a fixed distance x, a runner is considered bad at time t if

$$\|tv_i\| < x.$$

The expression $\|tv_i\|$ means the distance from tv_i to the nearest integer. Since the track is circular, integers represent the starting point again. So this measures how close runner i is to the start.

If we use the target distance

$$x = \frac{1}{n},$$

then one runner is bad when it lies within distance $\frac{1}{n}$ of the start. On the unit circle, this bad region has total length

$$\frac{1}{n} + \frac{1}{n} = \frac{2}{n}.$$

There are two pieces because a runner can be close to 0 from the right side or close to 1 from the left side. Since there are n runners, the average number of bad runners is

$$n \cdot \frac{2}{n} = 2.$$

If the full set V does not already satisfy

$$\kappa(V) \geq \frac{1}{n},$$

then the proof shows that there is a time when fewer than two runners are bad. Since the number of bad runners must be a whole number, "fewer than two" means at most one runner is bad. That one runner can be removed. Then all remaining runners are good at that time, so

$$\kappa(V \setminus \{v\}) \geq \frac{1}{n}.$$

This is the main idea behind making one runner invisible.

The paper also gives a stronger version when more than one runner is allowed to become invisible. If $s \geq 1$ runners are allowed to be removed, then the bound improves to

$$\kappa(V \setminus S) \geq \frac{s+1}{2n}.$$

Here, $S \subset V$ is the set of invisible runners, and

$$|S| = s$$

means that S contains exactly s runners. The notation $V \setminus S$ means the original speed set after removing all runners in S.

The number

$$\frac{s+1}{2n}$$

is the improved loneliness distance. It increases as s increases because removing more problematic runners makes it easier for the remaining runners to be far from the starting point.

This formula also matches the one-runner case. If $s = 1$, then

$$\frac{s+1}{2n} = \frac{1+1}{2n} = \frac{2}{2n} = \frac{1}{n}.$$

So the stronger formula includes the original invisible runner theorem as a special case.

The reason the expression $\frac{s+1}{2n}$ appears comes from the same averaging idea. If the bad distance is

$$x = \frac{s+1}{2n},$$

then one runner is bad for a total proportion

$$2x = 2\left(\frac{s+1}{2n}\right) = \frac{s+1}{n}$$

of the time. Since there are n runners, the average number of bad runners is

$$n \cdot \frac{s+1}{n} = s+1.$$

Therefore, there must be a time where at most s runners are bad. If we remove those s bad runners, then every remaining runner is at distance at least

$$\frac{s+1}{2n}$$

from the origin. This proves

$$\kappa(V \setminus S) \geq \frac{s+1}{2n}.$$

Overall, the invisible runner idea shows that even when the full runner set is difficult to handle, removing a small number of bad runners can force the remaining runners to satisfy a meaningful loneliness bound. The main tool is averaging: if the average number of bad runners is small enough, then there must be a moment when only a small number of runners are causing the problem.

Week 6 (June 24)

Recap

During this week's meeting, Dr. Ahmed Biniaz and I revisited *Case 1* of the Lonely Runner Conjecture and continued discussing *lacunary sequences*, focusing on how rapidly increasing runner speeds make it easier to guarantee the loneliness condition. Dr. Biniaz also introduced the *Lovász Local Lemma*, a probabilistic method used to show that certain bad events can be avoided even when dependencies exist between them. In addition, he encouraged me to continue making steady progress while taking time to carefully read mathematical papers and fully understand the ideas behind the proofs rather than rushing through them. For this week, he recommended that I focus primarily on carefully studying Ram Krishna Pandey's paper on the Lonely Runner Conjecture in order to strengthen my understanding of lacunary sequence arguments and related proof techniques.

Pandey's Lacunary Sequence Result

This week, I focused on Ram Krishna Pandey's paper *A Note on the Lonely Runner Conjecture*. The paper does not prove the full Lonely Runner Conjecture. Instead, it proves the conjecture for a special case where the runner speeds grow very quickly. These types of speeds are called *lacunary*, meaning there are large gaps between consecutive speeds.

Let

$$M = \{m_1, m_2, \ldots, m_n\}$$

be the set of runner speeds. Here, n is the number of runners, and $m_1, m_2, \ldots, m_n$ are their speeds. Pandey arranges the speeds in increasing order:

$$m_1 < m_2 < \cdots < m_n.$$

This ordering matters because the proof compares each speed to the speed directly before it.

Pandey assumes that for every consecutive pair of speeds,

$$\left(\frac{m_{j+1}}{m_j}\right)\left(\frac{n-1}{n+1}\right) \geq 2.$$

The ratio

$$\frac{m_{j+1}}{m_j}$$

measures how much faster runner $j+1$ is than runner j. The factor

$$\frac{n-1}{n+1}$$

comes from the length of the safe interval

$$\left[\frac{1}{n+1}, \frac{n}{n+1}\right].$$

That safe interval has length

$$\frac{n}{n+1} - \frac{1}{n+1} = \frac{n-1}{n+1}.$$

So Pandey's condition says that when the safe interval from one runner is stretched by the next speed, it becomes long enough to contain at least two units of length.

The condition can be rewritten as

$$\frac{m_{j+1}}{m_j} \geq 2\left(\frac{n+1}{n-1}\right).$$

This means the next speed must be more than double the previous speed, with a correction factor depending on n. For example, if $n = 3$,

$$\frac{m_{j+1}}{m_j} \geq 2\left(\frac{4}{2}\right) = 4.$$

So for 3 runners, each next speed must be at least 4 times the previous speed. If $n = 5$,

$$\frac{m_{j+1}}{m_j} \geq 2\left(\frac{6}{4}\right) = 3.$$

So for 5 runners, each next speed must be at least 3 times the previous speed. As n becomes large,

$$\frac{n+1}{n-1}$$

gets closer to 1, so the condition becomes closer to

$$m_{j+1} \geq 2m_j.$$

This explains why the theorem is about rapidly increasing speeds.

Main Idea of the Proof

Pandey's proof builds a sequence of nested intervals:

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_n.$$

Each interval I_j is a set of good times for runner j. The symbol

$$I_j$$

means the interval chosen for the j-th runner. The nesting condition means every new interval is placed inside the previous one.

For every time

$$x \in I_j,$$

Pandey wants

$$\|m_j x\| \geq \frac{1}{n+1}.$$

This means that at time x, runner j is at least distance

$$\frac{1}{n+1}$$

away from the nearest integer position on the circle. In other words, runner j is not too close to the starting point.

The reason nesting is useful is that if

$$I_1 \supset I_2 \supset \cdots \supset I_n,$$

then every time in the final interval I_n is also inside every earlier interval. Therefore, if

$$x \in I_n,$$

then x works for runner 1, runner 2, and all the way up to runner n. This is how Pandey finds one time that works for all runners simultaneously.

Step 1: Constructing the First Interval

Pandey starts with the slowest speed m_1. He defines

$$I_1 = \left[\frac{1}{m_1(n+1)}, \frac{n}{m_1(n+1)}\right].$$

This interval is chosen by working backwards from the safe region. The safe region for one runner's position is

$$\left[\frac{1}{n+1}, \frac{n}{n+1}\right].$$

Pandey wants $m_1 x$ to land inside this safe region. So he divides the endpoints by m_1. That gives

$$\frac{1}{m_1(n+1)} \quad \text{and} \quad \frac{n}{m_1(n+1)}.$$

That is where the formula for I_1 comes from.

If

$$x \in I_1,$$

then multiplying the whole interval by m_1 gives

$$m_1 x \in \left[\frac{1}{n+1}, \frac{n}{n+1}\right].$$

This means runner 1's position is safely away from the nearest integer. Therefore,

$$\|m_1 x\| \geq \frac{1}{n+1}.$$

The length of I_1 is found by subtracting the left endpoint from the right endpoint:

$$|I_1| = \frac{n}{m_1(n+1)} - \frac{1}{m_1(n+1)}.$$

Since the denominators are the same, the numerators subtract:

$$|I_1| = \frac{n-1}{m_1(n+1)}.$$

This can also be written as

$$|I_1| = \frac{1}{m_1}\left(\frac{n-1}{n+1}\right).$$

This form is useful because it separates the effect of the speed m_1 from the length of the safe region. The term

$$\frac{n-1}{n+1}$$

comes from the safe region, while the factor

$$\frac{1}{m_1}$$

comes from scaling time by the speed.

Step 2: The Inductive Step

Now suppose Pandey has already built an interval

$$I_{j-1} = [u_{j-1}, v_{j-1}].$$

Here, u_{j-1} is the left endpoint and v_{j-1} is the right endpoint. This interval already works for the earlier runners. The goal is to find a smaller interval

$$I_j = [u_j, v_j]$$

inside I_{j-1} that also works for runner j.

Pandey looks at what happens when the entire interval I_{j-1} is multiplied by the new speed m_j. The stretched interval is

$$[m_j u_{j-1}, m_j v_{j-1}].$$

Its length is

$$m_j v_{j-1} - m_j u_{j-1}.$$

Factoring out m_j, this becomes

$$m_j(v_{j-1} - u_{j-1}).$$

Since

$$v_{j-1} - u_{j-1} = |I_{j-1}|,$$

and by construction

$$|I_{j-1}| = \frac{1}{m_{j-1}} \left(\frac{n-1}{n+1} \right),$$

we get

$$m_j(v_{j-1} - u_{j-1}) = \frac{m_j}{m_{j-1}} \left(\frac{n-1}{n+1} \right).$$

This formula is important because it tells us how long the previous good interval becomes after being stretched by the new speed m_j.

By Pandey's assumption,

$$\frac{m_j}{m_{j-1}}\left(\frac{n-1}{n+1}\right) \geq 2.$$

So the stretched interval has length at least 2. This number 2 is used because any interval of length at least 2 is guaranteed to contain a complete unit interval between two consecutive integers.

Therefore, there exists an integer $\ell(j)$ such that

$$m_j u_{j-1} \leq \ell(j) < \ell(j) + 1 \leq m_j v_{j-1}.$$

This means the full unit interval

$$[\ell(j), \ell(j) + 1]$$

fits inside the stretched interval. The integer $\ell(j)$ is not a special fixed number. It simply represents whichever integer makes this complete unit interval fit inside the stretched interval.

Now divide everything by m_j:

$$u_{j-1} \leq \frac{\ell(j)}{m_j} < \frac{\ell(j)+1}{m_j} \leq v_{j-1}.$$

This brings the interval back from runner-position space into time space. Therefore,

$$\left[\frac{\ell(j)}{m_j}, \frac{\ell(j)+1}{m_j}\right] \subseteq I_{j-1}.$$

This is useful because inside this smaller time interval, runner j completes exactly one full lap.

Step 3: Keeping Only the Safe Part

Inside the interval

$$\left[\frac{\ell(j)}{m_j}, \frac{\ell(j)+1}{m_j}\right],$$

multiplying by m_j gives

$$[\ell(j), \ell(j) + 1].$$

This means runner j's position moves through one complete lap. However, the whole lap is not safe because the endpoints are integers, where the runner is exactly at the starting point.

So Pandey keeps only the middle safe part:

$$I_j = \left[\frac{\ell(j) + \frac{1}{n+1}}{m_j}, \frac{\ell(j) + \frac{n}{n+1}}{m_j}\right].$$

This formula comes from taking the safe region

$$\left[\frac{1}{n+1}, \frac{n}{n+1}\right]$$

and shifting it by the integer $\ell(j)$. The shifted safe region is

$$\left[\ell(j) + \frac{1}{n+1}, \ell(j) + \frac{n}{n+1}\right].$$

Then Pandey divides by m_j to convert it back into time. That is why both endpoints of I_j have denominator m_j.

For every

$$x \in I_j,$$

multiplying by m_j gives

$$m_j x \in \left[\ell(j) + \frac{1}{n+1}, \ell(j) + \frac{n}{n+1}\right].$$

The integer part $\ell(j)$ does not affect distance to the nearest integer. Therefore,

$$\|m_j x\| \geq \frac{1}{n+1}.$$

So I_j is a good interval for runner j.

The length of I_j is

$$|I_j| = \frac{\ell(j) + \frac{n}{n+1}}{m_j} - \frac{\ell(j) + \frac{1}{n+1}}{m_j}.$$

Since both terms have denominator m_j, subtracting gives

$$|I_j| = \frac{\left(\ell(j) + \frac{n}{n+1}\right) - \left(\ell(j) + \frac{1}{n+1}\right)}{m_j}.$$

The $\ell(j)$ terms cancel because shifting by a full integer does not change the length of the interval:

$$|I_j| = \frac{\frac{n}{n+1} - \frac{1}{n+1}}{m_j}.$$

Now subtract the fractions in the numerator:

$$|I_j| = \frac{\frac{n-1}{n+1}}{m_j}.$$

So

$$|I_j| = \frac{1}{m_j}\left(\frac{n-1}{n+1}\right).$$

This shows that I_j has the same type of length as the earlier intervals, but scaled by the new speed m_j.

Why This Proves the Result

The construction guarantees three things:

$$I_j \subseteq I_{j-1}.$$

This means the intervals stay nested, so no earlier work is lost.

$$|I_j| = \frac{1}{m_j}\left(\frac{n-1}{n+1}\right).$$

This keeps the interval length in the exact form needed to repeat the argument for the next speed.

$$\|m_j x\| \geq \frac{1}{n+1} \quad \text{for every } x \in I_j.$$

This means every time in I_j is good for runner j.

Repeating this process gives

$$I_1 \supset I_2 \supset \cdots \supset I_n.$$

Since I_n is inside all previous intervals, any time

$$x \in I_n$$

works for every runner. Therefore,

$$\|m_j x\| \geq \frac{1}{n+1}$$

for every

$$j = 1, 2, \ldots, n.$$

This proves the Lonely Runner Conjecture for speeds satisfying Pandey's lacunary condition.

Main Intuition

The main idea is that large speed gaps create room. A small interval that works for a slower runner becomes much longer when multiplied by a faster speed. If it becomes long enough to contain one full lap of the faster runner, then we can choose the safe middle part of that lap.

The number

$$\frac{n-1}{n+1}$$

comes from the length of the safe region in one lap. The number

$$2$$

is used because stretching the previous interval to length at least 2 guarantees that a complete unit interval fits inside it. The ratio

$$\frac{m_j}{m_{j-1}}$$

controls how much the next speed stretches the previous interval.

So the theorem's condition

$$\frac{m_j}{m_{j-1}} \left(\frac{n-1}{n+1} \right) \geq 2$$

is exactly the condition needed to keep the nested interval construction going.

Conclusion

Pandey's result shows that the Lonely Runner Conjecture is true when the speeds grow fast enough. The proof works by repeatedly choosing smaller safe time intervals. Each new interval is placed inside the previous one, so the final interval contains times that work for all runners. The large gaps between speeds guarantee that each new runner has enough room to complete a full lap inside the previous good interval, allowing Pandey to keep only the safe middle part.

Week 7 (June 30)

Recap

During this week's meeting, Dr. Ahmed Biniaz and I spent most of our discussion carefully going through Ram Krishna Pandey's paper on the Lonely Runner Conjecture, with a strong focus on fully understanding the proof rather than simply following the final result. In particular, we examined Pandey's interval construction argument, the inductive structure of the proof, and how rapidly increasing speed sequences allow the conjecture to hold under lacunary conditions. Dr. Biniaz emphasized the importance of understanding where each formula comes from, why each interval is constructed in a specific way, and the intuition behind the theorem rather than memorizing the algebraic steps. Toward the end of the meeting, he recommended that I continue studying another paper by Pandey related to the *bounded speeds* section discussed in the survey paper, specifically focusing on understanding the main theorems involving linearly bounded speed sets and attempting to work through some of the proofs independently in order to strengthen my mathematical intuition and overall understanding of current approaches to the conjecture.

Pandey's Bounded Speeds and Additive Structure Result

This week, I also began studying another paper by Ram Krishna Pandey, titled *On the Lonely Runner Conjecture*. Unlike Pandey's lacunary sequence paper, where the main idea was that the speeds grow very quickly, this paper studies special speed sets using additive structure. The main idea is that if the set of speeds has a strong arithmetic pattern, such as being contained in an arithmetic progression, then the Lonely Runner Conjecture can be verified for those speeds using elementary number theory.

Pandey begins by defining, for a finite set of speeds

$$M = \{m_1, m_2, \ldots, m_k\},$$

the quantity

$$\kappa(M) = \sup_{x \in (0,1)} \min_i \|xm_i\|.$$

Here, x represents time, m_i is the speed of the i-th runner, and xm_i is the position of that runner at time x, measured modulo 1. The expression

$$\|xm_i\|$$

means the distance from xm_i to the nearest integer, so it measures how far runner i is from the starting point. The inner expression

$$\min_i \|xm_i\|$$

takes the worst runner at time x, meaning the runner closest to the starting point. Then the outer expression

$$\sup_{x\in(0,1)}$$

means we choose the best possible time x. Therefore, $\kappa(M)$ is the largest distance that can be guaranteed for all runners in the speed set M at the same time.

In this notation, the Lonely Runner Conjecture says that if

$$|M| = k,$$

then

$$\kappa(M) \geq \frac{1}{k+1}.$$

This means that for k moving runners, there should exist a time when every runner is at least distance

$$\frac{1}{k+1}$$

from the starting point.

Equivalent Forms of $\kappa(M)$

Pandey then uses a result of Haralambis, which gives three equivalent ways to define $\kappa(M)$. The first one is

$$\kappa_1(M) = \sup_{x\in(0,1)} \min_i \|xm_i\|.$$

This is the standard continuous-time version of the Lonely Runner problem. It looks at all real times $x \in (0, 1)$.

The second version is

$$\kappa_2(M) = \sup_{(c,m)=1} \frac{1}{m} \min_i |cm_i|_m.$$

Here, m is a modulus, c is an integer relatively prime to m, and

$$|cm_i|_m$$

means the absolute least remainder of cm_i modulo m. This version changes the problem from real times to modular arithmetic. Instead of directly choosing a real time x, we choose a rational-type time related to

$$\frac{c}{m}.$$

The third version is

$$\kappa_3(M) = \max_{m=m_j+m_l} \frac{1}{m} \min_i |xm_i|_m, \qquad 1 \le x \le \frac{m}{2}.$$

This form is useful because it allows Pandey to prove lower bounds for $\kappa(M)$ by choosing a specific modulus m and a specific integer x. The paper states that

$$\kappa_1(M) = \kappa_2(M) = \kappa_3(M),$$

so Pandey is allowed to use whichever version is easiest for the proof.

Two Basic Properties

Pandey also uses two simple but important properties. First, if d is a positive integer and

$$dM_1 = M_2,$$

then

$$\kappa(M_1) = \kappa(M_2).$$

This means multiplying every speed by the same positive integer does not change the loneliness value. Intuitively, scaling all speeds together only changes how fast the same pattern repeats; it does not change the best possible guaranteed distance.

Second, if

$$M_1 \subset M_2,$$

then

$$\kappa(M_1) \ge \kappa(M_2).$$

This means adding more runners can only make the problem harder. If a time works for a larger set of runners, then it also works for any smaller subset. Therefore, the guaranteed distance for a smaller set cannot be worse than the guaranteed distance for a larger set.

Arithmetic Progressions

A major object in the paper is an arithmetic progression:

$$M = \{a, a+d, a+2d, \ldots, a+(k-1)d\}.$$

Here, a is the first speed, d is the common difference, and k is the number of speeds. This means each speed is obtained by adding d to the previous speed. For example, if $a = 3$, $d = 2$, and $k = 4$, then

$$M = \{3, 5, 7, 9\}.$$

Pandey proves a lower bound for $\kappa(M)$ when M is an arithmetic progression.

Theorem 3.1

Pandey states that if

$$M = \{a, a+d, a+2d, \ldots, a+(k-1)d\},$$

then

$$\kappa(M) > \frac{2a+(k-1)(d-1)}{2\{2a+(k-1)d\}}$$

when d is odd, and

$$\kappa(M) = \frac{1}{2}$$

when d is even.

The case where d is even is simpler. If d is even and a is odd, then all elements of M are odd. Choosing

$$x = \frac{1}{2}$$

gives

$$xm_i = \frac{m_i}{2}.$$

Since each m_i is odd, $\frac{m_i}{2}$ is always exactly halfway between two integers. Therefore,

$$\|xm_i\| = \frac{1}{2}.$$

This is the largest possible distance from an integer, so

$$\kappa(M) = \frac{1}{2}.$$

For the odd d case, Pandey chooses the modulus

$$m = 2a + (k-1)d.$$

This number comes from adding the first and last terms of the arithmetic progression:

$$a + \big(a + (k-1)d\big) = 2a + (k-1)d.$$

This choice is useful because the arithmetic progression becomes symmetric around

$$\frac{m}{2}.$$

Since

$$\gcd(d, m) = 1,$$

there exists an integer x such that

$$dx \equiv 1 \pmod{m}.$$

This means x is the modular inverse of d modulo m. Pandey uses this so that multiplying the arithmetic progression by x makes the common difference behave like 1 modulo m.

The proof then shows that

$$(a + ld)x \equiv \frac{m}{2} + \left(l - \frac{k-1}{2}\right) \pmod{m}.$$

This formula means that after multiplying by x, the elements of the arithmetic progression are placed near the middle of the circle, around $m/2$, instead of near 0. This is useful because being near $m/2$ means being far from multiples of m, which corresponds to being far from the starting point.

Using the $\kappa_3(M)$ form, Pandey obtains

$$\kappa(M) > \frac{2a + (k-1)(d-1)}{2\{2a + (k-1)d\}}.$$

The denominator

$$2\{2a + (k-1)d\}$$

comes from twice the modulus m, since

$$m = 2a + (k-1)d.$$

The numerator

$$2a + (k-1)(d-1)$$

comes from measuring how far the shifted elements stay away from 0 modulo m. This lower bound tells us that arithmetic progressions have enough regular structure to guarantee a positive loneliness distance.

Small Doubling and Freiman's Theorem

Pandey then studies sets M using the sumset

$$M + M.$$

This means

$$M + M = \{x + y : x \in M,\ y \in M\}.$$

The size

$$|M + M|$$

measures how additive-structured the set M is. If $|M + M|$ is small, then M behaves similarly to an arithmetic progression.

Pandey uses Freiman's theorem, which says that if

$$|M| = k, \qquad k \geq 4,$$

and

$$|M + M| = 2k - 1 + b < 3k - 3,$$

then M is contained in an arithmetic progression of length at most

$$k + b.$$

The number $2k - 1$ is the smallest possible size of $M + M$, which happens for arithmetic progressions. Therefore, writing

$$|M + M| = 2k - 1 + b$$

means b measures how far the set is from being a perfect arithmetic progression. Small b means M is very structured.

Theorem 3.2

Pandey's Theorem 3.2 says that if

$$|M| = k, \qquad k \geq 4,$$

and

$$|M+M| = 2k-1+b < 3k-3,$$

then

$$\kappa(M) > \frac{1}{k+1},$$

provided that if M is contained in an arithmetic progression with difference 1, the first term of that progression is greater than 1.

The proof begins by applying Freiman's theorem. Since M has small doubling, it is contained in an arithmetic progression

$$M \subseteq \{a, a+d, \ldots, a+(k+b-1)d\}.$$

This larger arithmetic progression has length $k+b$. Pandey then uses the fact that if

$$M \subseteq A,$$

then

$$\kappa(M) \geq \kappa(A).$$

This is because A has at least as many speeds as M, so proving a lower bound for the larger set A also gives a lower bound for M.

Using Theorem 3.1 on the larger arithmetic progression gives

$$\kappa(M) > \frac{2a+(k+b-1)(d-1)}{2\{2a+(k+b-1)d\}}.$$

Pandey then needs to show that this expression is greater than

$$\frac{1}{k+1}.$$

So he checks the inequality

$$\frac{2a + (k+b-1)(d-1)}{2\{2a + (k+b-1)d\}} > \frac{1}{k+1}.$$

Cross-multiplying gives

$$(k+1)\{2a + (k+b-1)(d-1)\} > 2\{2a + (k+b-1)d\}.$$

This step is just comparing two fractions. Since the denominators are positive, cross-multiplication is valid.

After simplifying, Pandey obtains

$$2a(k-1) > (k+b-1)\{k+1-(k-1)d\}.$$

This inequality is automatically true when

$$d \geq 2,$$

because the expression

$$k+1-(k-1)d$$

becomes non-positive or small enough that the left side is clearly larger.

The only delicate case is

$$d = 1.$$

When $d = 1$, the inequality becomes

$$2a(k-1) > 2(k+b-1).$$

Dividing by 2, this becomes

$$a(k-1) > k+b-1.$$

Using the assumption

$$2k - 1 + b < 3k - 3,$$

we get

$$k + b - 1 < 2k - 3.$$

So it is enough to have

$$a(k-1) > 2k - 3.$$

This is equivalent to

$$a > \frac{2k-3}{k-1}.$$

Since

$$\frac{2k-3}{k-1} < 2,$$

it is enough to assume

$$a > 1.$$

This explains the condition in the theorem: if the arithmetic progression has difference 1, its first term must be greater than 1.

Summary and Conclusion

Overall, this paper introduced a different approach to studying the Lonely Runner Conjecture by focusing on the *arithmetic structure* of the speed set rather than rapidly increasing speeds. The main idea is that if the speeds have enough regularity, then tools from additive number theory can be used to prove the required loneliness bound.

In **Theorem 3.1**, Pandey proves a lower bound for arithmetic progressions. The proof uses modular arithmetic by choosing a carefully constructed modulus and modular inverse to rearrange the speeds around the point opposite the starting position, allowing him to guarantee a minimum loneliness distance.

In **Theorem 3.2**, Pandey extends this result to more general speed sets. Using Freiman's Theorem, he shows that if the sumset $M + M$ is sufficiently small, then the speed set is contained inside a larger arithmetic progression. He then applies Theorem 3.1 to this larger progression and shows that the resulting lower bound is always greater than

$$\frac{1}{k+1},$$

thereby proving the Lonely Runner Conjecture for this class of structured speed sets.

Studying this paper has given me a much stronger understanding of arithmetic progressions, modular arithmetic, Freiman's Theorem, and additive combinatorics, and how they can be combined to prove special cases of the Lonely Runner Conjecture. I am currently continuing to study the remainder of the paper, with my next focus being a deeper understanding of **Theorem 3.3**.

Week 8 (July 7)

Recap

During this week's meeting, Dr. Ahmed Biniaz and I reviewed Pandey's Theorems 3.1 and 3.2 in greater depth, focusing on the main ideas behind the proofs and the role of additive structure. A major discussion point was the open case where the first term of an arithmetic progression is $a = 1$, with the long-term goal of understanding speed sets whose maximum speed is at most $2n$ and the first speed being 1. Dr. Biniaz also recommended that I study Section 8.2 of the Lonely Runner survey and develop a surface-level understanding of Tao's Theorems 22 and 23, as these results provide additional techniques and context that may be useful for extending the current approach. Over the coming week, my primary objective is to strengthen my understanding of these results and continue exploring how they relate to the $a = 1$ case. He also recommended I use my software to visualize the sets on a graph.

Tao's Theorem 22 (Linearly Bounded Speeds)

Dr. Biniaz recommended that I develop a general understanding of Tao's Theorem 22 rather than studying every technical detail of the proof. In Tao's paper, this result appears as *Proposition 1.5.* The theorem states that if an n-element speed set

$$V = \{v_1, v_2, \ldots, v_n\}$$

has largest speed

$$v_n < 1.2n,$$

then

$$\kappa(V) \geq \frac{1}{n+1},$$

meaning that the Lonely Runner Conjecture is true for every speed set satisfying this bound.

The key observation is that when every speed is at most approximately n, the speeds cannot be spread out arbitrarily. Instead, they become highly structured, allowing Tao to use number-theoretic arguments instead of treating the speeds as completely random. This makes the problem much easier than the general case.

The proof is by contradiction. Tao begins by assuming that the conjecture is *false* for such a speed set, meaning

$$\delta(v_1, \ldots, v_n) < \frac{1}{n+1}.$$

If this were true, then for *every* time t, at least one runner would always be within distance

$$\frac{1}{n+1}$$

of the starting point. In other words, there is never a time when all runners are simultaneously lonely.

To exploit this assumption, Tao proves an important intermediate result called **Lemma 5.1**. This lemma shows that if the conjecture fails, then the speed set is forced to contain many specific integers or multiples of integers. For example:

- for every integer j, some speed must be divisible by j;
- for many values of j, the speed j itself must actually appear in the speed set;
- in other ranges, either j or $2j$ must appear.

These conditions are obtained by carefully choosing special times such as

$$t = \frac{1}{j}$$

and nearby rational times. Since the assumption says some runner must always be close to the starting point, substituting these carefully chosen times forces the speeds to satisfy many divisibility and congruence conditions.

Tao then divides the possible values of j into several ranges. In each range, Lemma 5.1 guarantees that another distinct speed must belong to the set. Eventually, these forced values become too numerous. The argument ends by showing that the speed set would have to contain at least

$$n+1$$

distinct speeds, even though the set contains only

$$n$$

speeds. This contradiction proves that the original assumption was impossible, so there must exist a time when every runner is at least

$$\frac{1}{n+1}$$

away from the starting point.

After proving the theorem, Tao discusses an important open question. He asks whether the constant

$$1.2$$

can be replaced by

2.

This would extend the theorem to every speed set satisfying

$$v_n < 2n,$$

which would include important arithmetic progression examples such as

$$\{1, 2, \ldots, n\},$$

along with many other structured speed sets. These examples are believed to be among the most difficult cases of the Lonely Runner Conjecture. Tao leaves this as an open problem, and this question later motivated the work of Bohman and Peng, who introduced the method of *coprime mappings* in an attempt to extend the bound closer to $2n$.

Theorem 23 (Bohman and Peng)

After Tao proved the Lonely Runner Conjecture for speed sets with largest speed less than $1.2n$, he asked whether this bound could eventually be improved to $2n$. Bohman and Peng made one of the biggest advances toward answering this question.

Theorem 23 states that for sufficiently large n, if the largest speed satisfies

$$n < v_n \leq 2n - \exp\bigl(c(\log\log n)^2\bigr),$$

then

$$\kappa(V) \geq \frac{1}{n+1},$$

so the Lonely Runner Conjecture holds for this much larger family of bounded speed sets.

Unlike Tao's proof, which relies heavily on additive combinatorics, Bohman and Peng introduce a new technique called *coprime mappings*. The main idea is to pair numbers from different intervals so that each pair is relatively prime. Since coprime numbers do not share common prime factors, they behave more independently modulo different integers. This makes it much easier to find a time when all runners are simultaneously far from the starting point.

At a high level, the proof first divides the possible speeds into several intervals. Using graph-theoretic matching arguments, Bohman and Peng construct a pairing between these intervals where each matched pair is coprime. They then use these coprime relationships to carefully choose a time t so that the runners become well distributed around the unit circle. The coprime structure prevents the runners from clustering together, making it impossible for every runner to remain inside the bad interval at the same time. Therefore, there must exist a time when every runner is at least

$$\frac{1}{n+1}$$

away from the starting point.

The important takeaway is that Bohman and Peng extended Tao's bound from $1.2n$ to almost $2n$. Although the theorem still falls slightly short of the full bound $2n$, it demonstrates that the Lonely Runner Conjecture is true for speed sets whose largest speed is extremely close to $2n$. This represents one of the strongest known results for linearly bounded speed sets.

Shortly afterward, Carl Pomerance improved one of the key ingredients used in Bohman and Peng's proof. He showed that the required interval size for constructing coprime matchings could be reduced from approximately

$$\exp\big(c(\log\log n)^2\big)$$

to the much smaller quantity

$$c(\log n)^2.$$

As a result, the error term in Theorem 23 becomes even smaller, bringing the upper bound on v_n even closer to the desired value of $2n$. However, reaching the exact bound

$$v_n \leq 2n$$

remains an open problem. According to the survey, further improvements would require new progress on the *Jacobsthal problem*, which is closely connected to constructing these coprime matchings.

Connection to Pandey's Arithmetic Progression Results

Before exploring the current research direction, it is helpful to summarize the main ideas from Week 7, since they provide the mathematical foundation for the work that follows.

During the previous week, I studied Pandey's Theorems 3.1 and 3.2, which establish the Lonely Runner Conjecture for several families of arithmetic progression speed sets. The central idea behind these results is that arithmetic progressions possess a high degree of additive structure. Because the speeds follow a regular pattern, tools from modular arithmetic, additive combinatorics, and Freiman's Theorem can be used to analyze how the runners are distributed around the circle and prove the loneliness bound

$$\kappa(V) \geq \frac{1}{n+1}.$$

In Theorem 3.1, Pandey considers speed sets that are exact arithmetic progressions and derives an explicit lower bound for the loneliness distance by constructing a carefully chosen modular transformation that spreads the runners evenly around the circle. In Theorem 3.2, this idea is extended to more general speed sets by using Freiman's Theorem. If a speed set has sufficiently small doubling, then it must lie inside a short arithmetic progression, allowing the lower bound from Theorem 3.1 to be applied to the larger progression and, consequently, to the original speed set.

One important assumption in Pandey's proof is that when the arithmetic progression has common difference $d = 1$, the first term must satisfy

$$a > 1.$$

Therefore, the case where the first term is

$$a = 1$$

is not covered directly by his argument. This remaining case naturally motivates the next stage of the project and leads directly to the research direction discussed in the following subsection.

Research Idea: The Case $v_1 = 1$ and $v_n \le 2n$

One of the main open questions posed by Terence Tao is whether the Lonely Runner Conjecture holds whenever the largest speed satisfies

$$v_n \le 2n.$$

Since the speeds are ordered,

$$1 \le v_1 < v_2 < \cdots < v_n \le 2n,$$

this is often stated as *all speeds are at most* $2n$. My research focuses on an even more specific case by fixing the smallest speed to be

$$v_1 = 1.$$

The goal is to prove that every speed set

$$V = \{1, v_2, \ldots, v_n\}, \qquad v_n \le 2n,$$

satisfies

$$\kappa(V) \ge \frac{1}{n+1},$$

meaning there exists a time when every runner is at least $\frac{1}{n+1}$ away from the starting point.

The reason this case is interesting is that fixing the first speed to 1 gives the speed set additional arithmetic structure. Since every speed must lie between 1 and $2n$, there are only $2n$ possible speed values. As a result, the speeds cannot be completely arbitrary and must share arithmetic relationships such as small differences, common divisors, coprime pairs, or arithmetic progression structure. These are exactly the types of properties that previous researchers, including Pandey, Tao, and Bohman–Peng, have successfully used in their proofs.

Although a proof is not yet known, one possible idea is:

- Fix $v_1 = 1$ to normalize the problem and provide additional structure.
- Use the fact that $v_n \leq 2n$ to show that the speeds cannot behave randomly and instead must satisfy useful additive or number-theoretic properties.
- Exploit this structure, using ideas such as arithmetic progressions, coprime mappings, additive combinatorics, or modular arguments, to find a time t when every runner is simultaneously outside its bad interval.
- Conclude that
$$\kappa(V) \geq \frac{1}{n+1}.$$

The main challenge is that the bad intervals of different runners may overlap. The goal is to show that, because the speeds are so structured, these overlaps can never cover every possible time. Therefore, there should always exist at least one time when every runner is simultaneously outside its bad interval, meaning every runner is lonely. Considering I have no post graduate math experience, this may be tricky to implement in practice.

Week 9 (July 15)

Recap

During this week's meeting, which was rescheduled to Wednesday, Dr. Ahmed Biniaz and I discussed several improvements to my Lonely Runner Conjecture visualization software and the next direction of my research. We identified performance issues with the current implementation and planned several enhancements, including representing runners using vectors or line segments instead of triangles, displaying the loneliness threshold of $\frac{1}{n+1}$ as a horizontal reference line, adding a zoom feature, allowing individual runner speeds to be modified dynamically, and improving the visualization of point coordinates and the best loneliness distance over different times. On the theoretical side, Dr. Biniaz explained that Ram Krishna Pandey's results cannot handle the case where the first term $a = 1$, even beyond arithmetic progressions, making his theorem less useful for our overall research goal. For next week, I will study Pomerance's improvement in depth, focusing on how the error term was reduced from $\exp\big(c(\log\log n)^2\big)$ to approximately $c(\log n)^2$, bringing the bound closer to $2n$.

Software Implementation Updates

This week, I continued developing the Lonely Runner Conjecture visualization software by adding several new features and improving the original implementation. The initial version of the program accepted the number of runners and their speeds as input, then plotted each runner's distance function as a sequence of triangular graphs between 0 and $\frac{1}{2}$. It also displayed the lower and upper reference lines of the graph, allowing the distance functions for different speed sets to be visualized.

The first major addition was the loneliness threshold,

$$\frac{1}{n+1},$$

which is now drawn as a horizontal dashed line across the graph. This line is automatically calculated from the input value of n each time the graph is generated. A label displaying $\frac{1}{n+1}$ was also added beside the graph so that the loneliness bound is clearly visible while examining the runner distance functions.

I also extended the program so that it no longer only draws the graphs, but also analyzes them. As each triangle is created, the program stores every rising and falling edge as an individual line segment. After all runners have been drawn, every pair of line segments from different runners is compared to determine whether they intersect. A geometric line-segment intersection algorithm was implemented to calculate the coordinates of each intersection point.

Not every intersection is displayed on the graph. Two additional conditions were added to filter the results. First, an intersection must lie at or above the loneliness threshold $\frac{1}{n+1}$. Second, the program checks whether another runner's distance graph lies below that intersection at the same time. This is done by evaluating every runner's distance function at the intersection's x-coordinate. If another graph exists below the intersection, the point is discarded. As a result, only intersections that satisfy both conditions are shown on the graph.

Several helper functions were added to support these new features. One function computes the intersection point of two line segments, another evaluates a runner's distance from the starting point at any given time, and a third determines whether the vertical path beneath an intersection is clear of every other runner's graph. Together, these functions perform the calculations needed to determine which intersection points should be displayed.

The program was also updated to perform additional input validation by checking that the number of speeds entered matches the specified value of n and that all speeds are positive integers. A small numerical tolerance was introduced when comparing floating-point values to avoid precision errors, and duplicate intersection points are removed before being plotted. Finally, the plotting area was adjusted to leave space for the loneliness label outside the graph while preserving the original visualization of the runner distance functions.

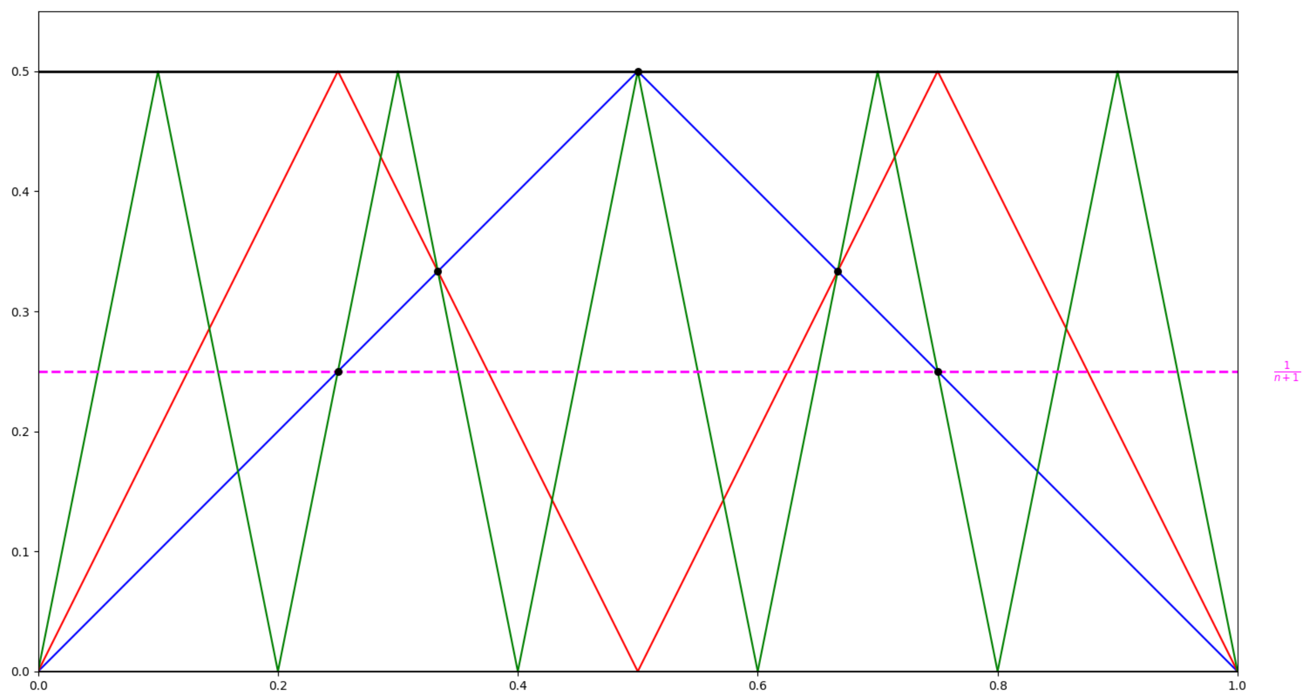


Figure 6: Updated software implementation with $\frac{1}{n+1}$ horizontal line, and important points of intersection. This is a sample run where n = 3, and the values in the speed set are 1, 3, and 5.

Introduction to Coprime Matchings

The paper studies a matching problem involving two intervals of positive integers. An interval of integers means a set of consecutive numbers. For example,

$$I = \{2, 3, 4\} \qquad \text{and} \qquad J = \{8, 9, 10\}$$

are both intervals of length 3.

Suppose that two intervals I and J have the same number of elements. A matching pairs every number in I with exactly one number in J. In mathematical language, this is called a bijection. A bijection means that every number in I is matched with one different number in J, and every number in J is used exactly once.

The goal is to find a *coprime matching*. Two positive integers are coprime if their greatest common divisor is 1. This means that they do not share any common factor larger than 1. For example,

$$\gcd(4, 9) = 1,$$

so 4 and 9 are coprime. However,

$$\gcd(4, 10) = 2,$$

so 4 and 10 are not coprime.

Therefore, a coprime matching between I and J is a pairing in which every matched pair has greatest common divisor equal to 1. If $f : I \to J$ represents the matching, then the condition is

$$\gcd(i, f(i)) = 1$$

for every $i \in I$.

A coprime matching does not always exist. One obstruction occurs when one number in an interval shares a common factor with every number in the other interval. For example, suppose $1 \notin I$ and J contains a number j that is divisible by the product of all the numbers in I. Then j is divisible by every member of I, so it cannot be matched coprimely with any number in I.

Another obstruction can occur when both intervals contain too many even numbers. Any two even numbers have a common factor of 2, so they cannot form a coprime pair. For example, consider

$$I = \{2, 3, 4\} \qquad \text{and} \qquad J = \{8, 9, 10\}.$$

The interval I contains the even numbers 2 and 4, while J contains the even numbers 8 and 10. Since both intervals contain more even numbers than odd numbers, at least one even number must be matched with another even number. That pair will have greatest common divisor at least 2, so a coprime matching is impossible.

To avoid this problem, the authors require the common interval length to be even. They write the length as

$$2m.$$

An interval of even length contains exactly the same number of even and odd integers. Therefore, the problem caused by both intervals having a strict majority of even numbers cannot occur.

The paper also requires the numbers in the intervals not to be too large compared with the interval length. The intervals are assumed to lie inside

$$[n] = \{1, 2, \ldots, n\}.$$

This means that every number in both intervals is at most n.

The main result states that there is a positive constant c such that a coprime matching exists when the two intervals have the same even length and their common length is greater than

$$c(\log n)^2.$$

Here, $\log n$ is the logarithm of n, and c is a fixed positive constant. The exact value of c is less important than the general meaning of the condition. It says that the intervals must be sufficiently long compared with the size of the largest possible number n.

In simple terms, the theorem says that if two intervals are contained in $\{1, 2, \ldots, n\}$, have the same even length, and are long enough, then their numbers can be paired so that every matched pair is coprime.

The paper improves an earlier result of Bohman and Peng. Their work also studied conditions that guarantee coprime matchings between intervals. The new paper lowers the required interval length to approximately

$$c(\log n)^2.$$

This means that the result applies to shorter intervals than the earlier theorem.

The introduction also mentions several previous results. One result states that if one interval is

$$[n] = \{1, 2, \ldots, n\}$$

and the other is any interval of length n, then a coprime matching exists. The presence of the number 1 is especially useful because

$$\gcd(1, a) = 1$$

for every positive integer a. Therefore, 1 can be matched with any number.

Bohman and Peng also proved that if I and J are intervals of the same length $k \geq 4$ and the number k belongs to I, then a coprime matching exists. This proved an earlier conjecture about coprime matchings of intervals.

The main purpose of the introduction is to define what a coprime matching is, explain why such matchings do not always exist, and motivate the conditions needed to guarantee one exists. It introduces the main obstacles, such as numbers sharing common factors or intervals containing too many even numbers, before showing how these issues can be avoided by restricting the interval length and size of the numbers. Finally, it summarizes previous results in the area and explains how the new theorem improves upon earlier work by proving that coprime matchings exist under weaker conditions.

Week 10 (July 22)

Recap

This week's meeting was cancelled because Dr. Ahmed Biniaz was feeling unwell with a cold and a sore throat. The meeting has been rescheduled for next Tuesday. In the meantime, I will continue improving the Lonely Runner Conjecture visualization software by refining its implementation and adding new features where appropriate. I will also continue studying the paper on coprime matchings to strengthen my understanding of the underlying ideas and their connection to the Lonely Runner Conjecture.

Program Improvements

The original program plotted each runner's distance from the starting point using the function

$$\|tv_i\|,$$

where t is the time and v_i is the speed of runner i. Several improvements were added to make the graph identify the most important times more accurately.

First, the program now finds intersections between the runner graphs and only keeps intersections that are at or above the Lonely Runner bound

$$\frac{1}{n+1}.$$

These intersections represent times when at least two runners have the same distance from the starting point.

The definition of a valid point of importance was also improved. Originally, the program checked whether a vertical line could be drawn downward from an intersection without crossing another runner's graph. However, runners located exactly at distance 0 were ignored. This caused some invalid points to be marked even though another runner was at the starting point at that time. The program now rejects an intersection whenever any runner is below it, including runners at distance 0. Therefore, a point is marked only when

$$\|tv_i\| \geq y$$

for every runner, where y is the height of the intersection.

The valid intersections are displayed as black points. Among these points, the program finds the greatest loneliness value, which is the largest height reached by a valid point. This value represents the largest minimum distance that all runners have from the starting point at the same time.

If the greatest loneliness value is strictly larger than

$$\frac{1}{n+1},$$

the program draws an orange dotted horizontal line at that height. It also estimates the value as both a decimal and a fraction. For example, a value close to 0.3333 is displayed as approximately 1/3. The fraction is produced using Python's `Fraction` class with a limited denominator.

A legend box was added above the graph so that the labels do not overlap the runner lines. The pink dashed line is identified as the Lonely Runner bound $1/(n+1)$, while the orange dotted line is identified as the greatest loneliness value. If the greatest loneliness value is exactly $1/(n+1)$, the program reports this in the legend but does not draw a second line because the pink line already represents that value. If no valid point exists, the greatest loneliness value is shown as `N/A`.

Finally, pressing the reset button now clears the entries and graph and returns both legend values to `N/A`. These changes make the program more mathematically accurate and make the important values easier to understand from the graph.

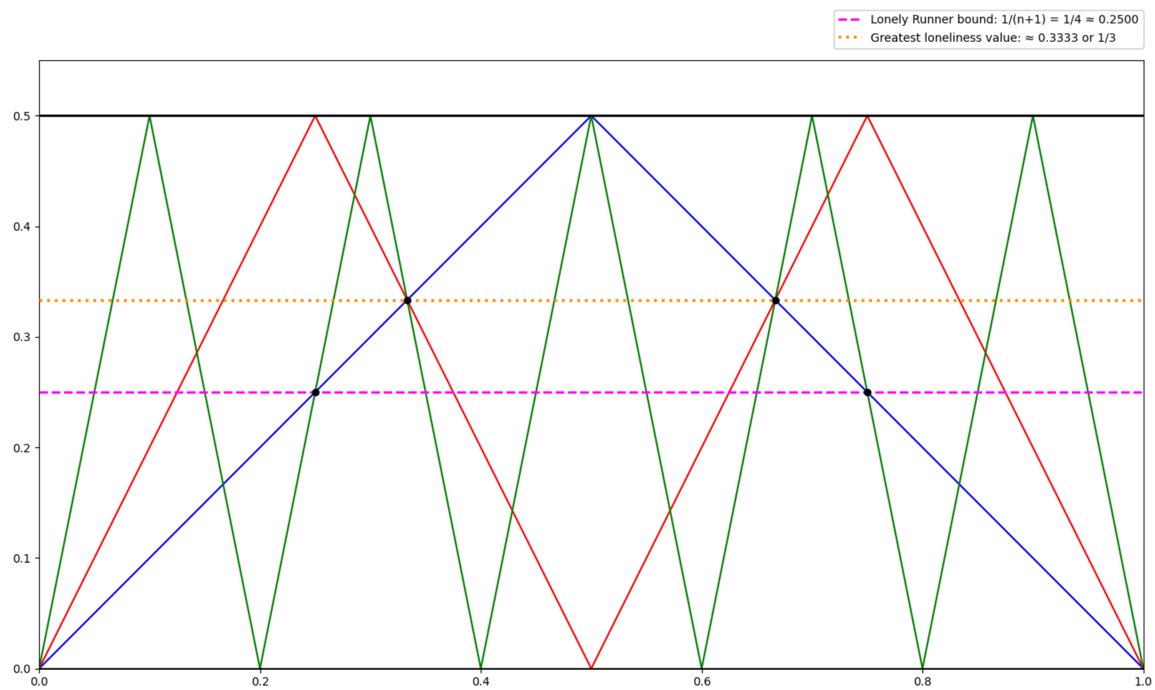


Figure 7: Updated software example when n = 3 and the values are 1, 2 and 5. Identifies valid points of intersection, computes the greatest loneliness value, and displays both it and the bound $\frac{1}{n+1}$ for easy comparison.

The Set-Up for Coprime Matchings

The purpose of this section is to turn the coprime matching problem into a graph problem. The authors then use Hall's Marriage Theorem to prove that a complete matching exists.

Hall's Marriage Theorem

Let I and J be two finite sets with the same size:

$$|I| = |J| = m.$$

A matching pairs elements of I with different elements of J. A perfect matching uses every element of both sets exactly once.

Hall's Marriage Theorem says that a perfect matching exists if every subset $S \subseteq I$ has at least $|S|$ possible partners in J.

The paper uses the following equivalent version.

Lemma 1. A perfect matching exists if, for every

$$S \subseteq I \qquad \text{and} \qquad T \subseteq J$$

with

$$|S| + |T| > m,$$

there is at least one allowed pair

$$(s, t), \qquad s \in S,\ t \in T.$$

In simple terms, two large subsets cannot be completely disconnected. If together they contain more than m elements, then at least one element of S must be able to match with one element of T.

Why this is related to Hall's theorem. Suppose S does not have enough neighbours in J. If $N(S)$ is the set of possible partners of S, then

$$|N(S)| < |S|.$$

Let

$$T = J \setminus N(S).$$

Then T contains all elements of J that cannot match with any element of S. Therefore, there is no edge between S and T.

Also,

$$|S| + |T| = |S| + m - |N(S)| > m.$$

Thus, failing Hall's condition creates two large subsets with no allowed pair between them. The reverse argument also works, so the two conditions are equivalent.

Bipartite Graph Interpretation

The authors represent the problem using a bipartite graph.

The two sides of the graph are I and J. An edge is drawn between

$$s \in I \qquad \text{and} \qquad t \in J$$

whenever s and t are allowed to be matched.

Therefore, proving a perfect matching exists is the same as proving that this bipartite graph has a perfect matching.

2-Coprime Numbers

The authors first prove a slightly weaker result using 2-coprime numbers.

Two integers s and t are called 2-coprime if no odd prime divides both of them.

Equivalently,

$$\gcd(s,t)$$

is a power of 2, such as

$$1, 2, 4, 8, \ldots.$$

For example,

$$\gcd(6,8) = 2,$$

so 6 and 8 are 2-coprime.

However,

$$\gcd(6,15) = 3,$$

so 6 and 15 are not 2-coprime because they share the odd prime 3.

Ordinary coprime numbers must satisfy

$$\gcd(s,t) = 1.$$

Thus, coprime numbers are always 2-coprime, but 2-coprime numbers are not always coprime.

Proposition 1. There is a positive constant c such that, for sufficiently large n, the following is true.

Let I and J be arithmetic progressions of length m, contained in

$$[n] = \{1, 2, \ldots, n\},$$

with common difference 1 or 2. Assume

$$m > c(\log n)^2.$$

If

$$S \subseteq I, \qquad T \subseteq J,$$

are nonempty and satisfy

$$|S| + |T| \geq m,$$

then there exist

$$s \in S, \qquad t \in T$$

that are 2-coprime.

An arithmetic progression with common difference 1 is an ordinary interval:

$$a, a+1, a+2, \ldots.$$

An arithmetic progression with common difference 2 has the form

$$a, a+2, a+4, \ldots,$$

so all of its elements have the same parity.

The proposition says that two sufficiently large subsets cannot avoid every 2-coprime pair.

Corollary 1. Under the conditions of Proposition 1, there is a bijection between I and J such that corresponding elements are 2-coprime.

To prove this, create a bipartite graph where

$$s \in I$$

is connected to

$$t \in J$$

exactly when s and t are 2-coprime.

Proposition 1 guarantees that every sufficiently large pair of subsets contains an edge. Therefore, Lemma 1 gives a perfect matching.

This perfect matching is exactly a one-to-one 2-coprime pairing between I and J.

The next step is to turn a 2-coprime matching into a normal coprime matching.

Corollary 2. There is a coprime matching between two intervals I and J if either:

1. both intervals have length $2m$; or
2. both intervals have length $2m+1$, and their first elements have opposite parity.

Suppose first that both intervals have length $2m$.

Separate the numbers into even and odd parts:

$$I_0 = \{\text{even elements of } I\}, \qquad I_1 = \{\text{odd elements of } I\},$$

and similarly define

$$J_0, \qquad J_1.$$

Each interval has exactly m even numbers and m odd numbers.

Use Corollary 1 to match

$$I_0 \text{ with } J_1$$

and

$$I_1 \text{ with } J_0.$$

Every matched pair has opposite parity. Therefore, the two numbers cannot both be divisible by 2.

They are also 2-coprime, so they share no odd prime factor. Hence their greatest common divisor must be

$$1.$$

Therefore,

$$\text{2-coprime} + \text{opposite parity} \Longrightarrow \text{coprime}.$$

The same idea works for intervals of length $2m+1$ when their first elements have opposite parity. In that case, the numbers of even and odd elements still match correctly across the two intervals.

Corollary 3

Corollary 3 handles the remaining odd-length cases, where the two intervals begin with the same parity.

If both first elements are odd, the authors choose one special coprime pair and then use Corollary 2 to match the remaining elements.

If both first elements are even, they can still match all elements, but one pair may have

$$\gcd(s,t) = 2$$

instead of 1. Every other pair is coprime.

The intervals are split near their middle so that each remaining piece is still large enough for the earlier results to apply.

Main Idea

The entire set-up follows this chain:

$$\boxed{\text{Hall's theorem} \longrightarrow \text{2-coprime matching} \longrightarrow \text{match opposite parities} \longrightarrow \text{coprime matching.}}$$

The main idea is that Hall's theorem changes the problem from constructing every pair directly into proving that large subsets always contain at least one allowable pair. The parity argument then changes 2-coprime pairs into ordinary coprime pairs.

Week 11 (July 28)

Recap

During this week's meeting, we decided to shift our primary focus away from the coprime matching paper and instead concentrate on developing a geometric approach to the Lonely Runner Conjecture using our software implementation. The main objective is to study the case where the smallest speed is $v_1 = 1$ and the largest speed satisfies $v_n \leq 2n$, where n is the number of runners (or speeds). We discussed that when the smallest speed is greater than 1, there is a known geometric property: the graph of the first speed always intersects the graph of the largest speed at a point that represents a valid lonely time. However, it is still unknown whether a similar property holds when $v_1 = 1$. For the coming week, I will continue improving the software by recording the coordinates of all valid intersection points, adding different features, and labeling each graph with its corresponding speed. I will also investigate whether there exist examples with no valid lonely time occurring at an intersection involving speed 1. If no such counterexample can be found, the next step will be to look for a general geometric pattern that could be proved and potentially establish that a valid intersection with speed 1 must always exist.

Software Improvements

This week, several improvements were made to the visualization software to support the search for new geometric properties of the Lonely Runner Conjecture. The program was updated to automatically record and display the (x, y) coordinates of every valid point of intersection after all validity conditions are satisfied, allowing these points to be analyzed more systematically. The software also computes the greatest loneliness value among all valid intersections and displays it alongside the theoretical bound $\frac{1}{n+1}$, making it easy to compare the computed value with the conjectured minimum bound. Interactive zooming was implemented by adding mouse wheel controls, dedicated zoom in and zoom out buttons, and a reset zoom option, allowing specific regions of the graph to be examined in greater detail. To improve readability, each runner is now assigned a unique color, and the legend identifies both the runner number and its corresponding speed. Together, these improvements make the visualizations clearer, simplify the analysis of valid intersection points, and provide a stronger computational tool for identifying geometric patterns that may lead to a proof of the special case being investigated.

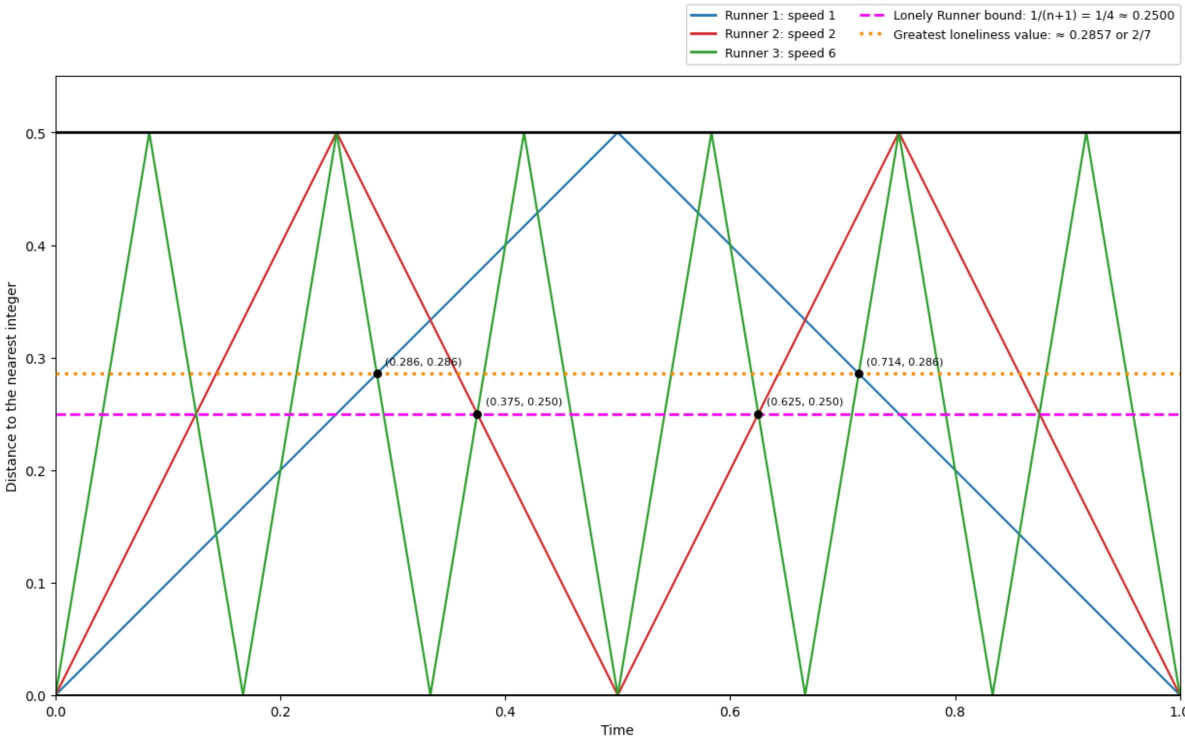


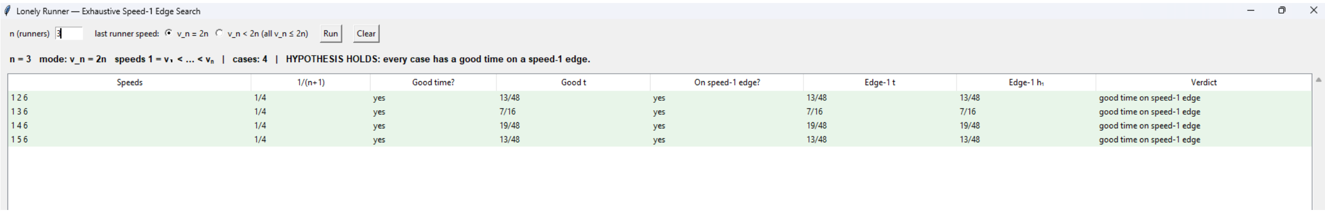
Figure 8: Updated visualization software displaying labeled runners, valid points of intersection, the greatest loneliness value, the bound $\frac{1}{n+1}$, and interactive zoom controls.

Special Case Search Program

To investigate the special case where the smallest speed is $v_1 = 1$ and the largest speed satisfies $v_n \leq 2n$, I developed a separate Python program that performs an exhaustive search. Instead of testing speed sets manually, the program automatically generates every increasing speed set satisfying either $v_n = 2n$ or $v_n \leq 2n$.

For each speed set, the program computes the loneliness bound $\frac{1}{n+1}$ and searches for a valid lonely time by examining all important event times where the runner graphs change. It then determines whether a valid lonely time exists and, more importantly, whether that lonely time occurs on the graph of the runner with speed 1, which is the property currently being investigated.

The program displays the results in a table showing the speeds, the loneliness bound, whether a valid lonely time exists, whether it occurs on the speed-1 graph, and the corresponding time. Finally, each case is classified as either satisfying the hypothesis, having a valid lonely time but not on the speed-1 graph, or being a counterexample to the Lonely Runner Conjecture. This allows hundreds of cases to be tested automatically and provides computational evidence to help identify patterns or possible counterexamples.



Speeds	1/(n+1)	Good time?	Good t	On speed-1 edge?	Edge-1 t	Edge-1 h₁	Verdict
1 2 6	1/4	yes	13/48	yes	13/48	13/48	good time on speed-1 edge
1 3 6	1/4	yes	7/16	yes	7/16	7/16	good time on speed-1 edge
1 4 6	1/4	yes	19/48	yes	19/48	19/48	good time on speed-1 edge
1 5 6	1/4	yes	13/48	yes	13/48	13/48	good time on speed-1 edge

Figure 9: Exhaustive search program that automatically tests all speed sets satisfying $v_1 = 1$ and $v_n \leq 2n$ (or $v_n = 2n$) to determine whether a valid lonely time exists on the graph of the runner with speed 1.

Searching for a Counterexample

During last week's meeting, Dr. Biniaz and I focused on investigating the special case $v_1 = 1$ and $v_n \leq 2n$. Our initial goal was to determine whether every speed set has a good lonely time that occurs at an intersection involving the speed-1 graph. To investigate this, I developed an exhaustive table program that tests every admissible speed set for a given n. Using this program, I found several candidate counterexamples to this stronger geometric hypothesis. One example is the speed set $\{1, 3, 5, 8\}$, where a lonely time exists, but no valid lonely intersection appears on the speed-1 graph. This does not contradict the Lonely Runner Conjecture itself, since a lonely time still exists; rather, it suggests that the stronger geometric property may not always hold.

Week 12 (August 4)

Recap

During this week's meeting, I presented several improvements to my Lonely Runner Conjecture graph software, including interactive zooming with the mouse wheel, displaying the mouse cursor coordinates, listing the coordinates of valid intersection points, identifying each runner by its corresponding speed and colour, and expanding the colour palette for improved visualization. I also introduced a new exhaustive table program that generates every admissible speed set for the special case $v_1 = 1$ with either $v_n = 2n$ or $v_n \leq 2n$. The program checks the loneliness bound, determines whether a lonely time exists, identifies the first valid lonely time, tests whether it occurs on the graph of Runner 1, and classifies each case accordingly.

Using the table program, we found several counterexamples to our original hypothesis that every lonely time would occur at an intersection on the speed-1 graph. Since these counterexamples exist, this stronger geometric property cannot be used directly to prove the special case. As a result, we shifted our focus to a new direction. We will now investigate whether every speed set with maximum speed at most $2n-3$ has a good intersection on either the speed-1 or speed-2 graph, noting that the case where speed 1 is absent is already known to hold. My supervisor will also be away for the next two weeks, and since my NSERC USRA contract ends on August 21, we plan to begin preparing a draft paper that summarizes the software developed, computational observations, candidate counterexamples, techniques explored, and the main ideas investigated throughout the project.

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